



OIL INDUSTRY SAFETY DIRECTORATE

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FIRE PROTECTION FACILITIES FOR PETROLEUM REFINERIES, PETROCHEMICAL COMPLEXES & OIL/GAS PROCESSING PLANTS OISD-STANDARD-116

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OIL INDUSTRY SAFETY DIRECTORATE
Government of India
Ministry of Petroleum & Natural Gas
8th Floor, OI DB Bhavan, Plot No. 2, Sector – 73, Noida – 201301 (U.P.)
Website: <https://www.oisd.gov.in> Tele: 0120-2593833

Oil Industry Safety Directorate

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PREFACE

Indian petroleum industry is the energy lifeline of the nation, and its continuous performance is essential for prosperity of the country. As the industry deals with inherently inflammable substances throughout its value chain – upstream, midstream and downstream – Safety is of paramount importance to this industry as safe performance at all times can ensure optimum return of these national assets and resources including sustainability.

To ensure proper implementation of the various aspects of safety in the oil & gas industry, the Government of India, vide Resolution No. R-13013/4/84-OR-1 dated 10th January 1986 set up a Safety Council at the apex, under the administrative control of the Ministry of Petroleum & Natural Gas (MoP&NG). To assist the Safety Council, a technical directorate, namely the Oil Industry Safety Directorate (OISD) under the aegis of MoP&NG, has been entrusted with the responsibility of formulating standards, overseeing its implementation through safety audits to enhance safety level in the oil & gas industry.

OISD has developed a rigorous, multi-layer, iterative and participative approach for development of standards. OISD develops and revises Standards, Guidelines & Recommended Practices for the oil and gas sector through a participative process involving all the stakeholders, drawing input from national and international standards and adapting them by leveraging the experience of the functional committee. These standards cover inbuilt design safety, asset integrity and best operating practices in the field of production, processing, storage and transport of petroleum.

The participative process followed in standard formulation has resulted in excellent level of compliance by the industry. It also goes to prove the old adage that self-regulation is the best regulation. The quality and relevance of OISD standards had been further endorsed by their adoption/ reference in various statutory rules and regulations of the land.

OISD and industry together strive to achieve nil incident in the entire hydrocarbon value chain. We at OISD are confident that the provisions of this standard, when implemented in totality, would go a long way in ensuring safe operation in the oil and gas industry.

Executive Director
Oil Industry Safety Directorate

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Wherever Acts/ Rules/ Regulation and National/ International Standards are mentioned in the standard, the same relates to in-vogue version of such documents. Whenever there is any difference between the statutory norm and OISD standard, more stringent requirement should be followed. However, in case of any difference in the requirement between OISD standards, the requirement as specified in the standard released at a later stage shall prevail.

The industry has to carry out gap analysis and make necessary changes in their system in line with the revised standard and the requirement should be implemented within a time frame of 1 year (from the date of release of revised standard).

Though every effort has been made to assure the accuracy and reliability of the information contained in these documents, OISD hereby expressly disclaims any liability or responsibility for loss or damage resulting from the use of OISD standard.

The standards would be reviewed periodically based on the decision in the Steering Committee/ Safety Council, industry request, technological development, recommendations of high-level committee and accident investigation reports. Generally, OISD standards are reviewed and revised, reaffirmed or withdrawn at least every ten years.

Suggestions for revision are invited from the oil and gas industry and should be submitted to OISD at ps-ed.oisd@gov.in.

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Amitabh Singh	GM (Fire Service)	ONGC
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Sandeep Kumar Sahu	Chief Manager (LPG-M&I)	IOCL
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Anshul Kumar Tiwari	Manager (PSM)	HMEL
Ramawatar Meena (Till 18 th June 2024)	Deputy DG of mines safety	DGMS
Muralidhar Bidari (From 19 th June 2024)	Director of mines safety	DGMS
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Amit Sharma	Joint Director	OISD

OISD recognize the support & cooperation of several other experts from industry, who contributed to the preparation, review and finalization of this document.

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1.0 INTRODUCTION

Petroleum refineries and petrochemical complexes play a crucial role in India's energy landscape. By virtue of basic properties of raw material, intermediates, finished products, and manufacturing processes, these industries pose a major risk of disaster like fire and explosion besides toxic hazard. In spite of sound engineering and operating procedures combined with competent personnel, there always remains a residual risk in day-to-day operation and maintenance of these units. This can pose a major threat to its employees, assets, and community at large. Most of the emergency situations can be effectively mitigated by well-designed fire protection system.

Refinery and Petrochemical complexes comprise of Process units, Storage tank farms, Control rooms, Utility production and distribution, Captive Power generation, Electrical sub-stations and power distribution network, workshops and warehouses, Laboratory, Non-plant buildings like Administration offices, Medical Center, Fire station and various other utility facilities. Every area or part of the entity shall be adequately protected commensurately with the risk profile and meeting the industry norms.

In general, the location of Refinery and Petrochemical complexes is generally away from the city limits and populated locality during phase of design. The requirement of firefighting facilities, described in this standard is based on the consideration that the firefighting services from city fire brigade and/or from other neighboring industries is not available as an immediate response. Hence industry to be self-sufficient in terms of emergency response infrastructure and resources to mitigate all credible and worst-case scenarios.

2.0 SCOPE

- i) This standard covers the minimum design criteria and details of the various fire protection facilities to be provided in petroleum refineries, petrochemical complexes, and oil/gas processing plants.
- ii) This standard also covers the facilities located within the refineries and oil/ gas processing plants which includes:
 - LPG storage, handling, and bottling plants
 - Petrochemical complexes
 - Marketing & Pipeline Terminals with their loading / un-loading/ pumping and handling facilities
- iii) This standard does not cover the facilities located outside the premises of Refineries & gas processing plants such as LPG mounded storage, handling & bottling plants, LNG storage Petroleum Depots, Terminals, Lube Blending plants & Pipeline Installations-
However, this standard shall be applicable to those of the above-mentioned facilities located outside the premises under the same management and in close proximity to the refineries & gas processing plants. For such cases, common firewater storage and pumping facilities should be considered.



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3.0 DEFINITIONS

3.1 AUTO IGNITION TEMPERATURE

A temperature of a substance is the lowest temperature in which it spontaneously ignites in a normal atmosphere without an external source of ignition

3.2 CLASSIFICATION OF PETROLEUM PRODUCTS

a) General Classification of Petroleum Products:

Petroleum products are classified according to their closed cup FLASH POINTS as follows:

Class 'A' Petroleum: Liquids having flash point below 23°C.

Class 'B' petroleum: Liquids having flash point of 23°C and above but below 65°C.

Class 'C' petroleum: Liquids having flash point of 65°C and above but below 93°C.

Excluded Petroleum: Liquids having flash point of 93°C and above.

Liquefied gases, including LPG do not fall under this classification but form a separate category.

b) Classification for heated products : At locations where the handling temperatures are higher than the flash point of the product in circumstances where product handled is artificially heated above its flash point, excluded product shall be considered as Class 'C' product, Class 'C' product shall be considered as Class 'B' product and Class 'B' product shall be considered as Class 'A' product.

3.3 CLASSIFICATION OF FIRE

Class A Fire: Involve combustible material, such as wood, paper, rubber, and many plastics, etc., where the cooling effect of water is essential for extinction of fire.

Class B Fire: It involves flammable liquids, petroleum products, or the like, where a blanketing effect is essential.

Class C Fire: Involve flammable gases under pressure including liquefied gases where it is necessary to inhibit the burning gas at a fast rate with an inert gas, powder, or vaporizing liquid for extinguishment.

Class D Fire: involve combustible metals such as magnesium, aluminum, zinc, sodium, potassium. The burning metals are reactive to water and water containing agents, and in certain cases carbon dioxide, halogenated hydrocarbons, and ordinary dry powders. These fires require special media and technique to extinguish.

Class F Fire: involve cooking oils and fats, or the like, where a wet chemical fire extinguisher is essential.

3.4 CLEAN AGENT

Electrically non-conductive, volatile, or gaseous fire extinguishants that do not leave a residue upon evaporation and meets the requirements given in the latest NFPA Standard 2001 (latest edition) on clean agent fire extinguishing system in line with environmental consideration of Kyoto & Montreal protocols and latest MoEFCC guidelines.



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3.5 CONTROL ROOM

“Control Room” is a protective enclosure equipped with control & safety systems and/ or environmental treatment necessary for proper functioning of the Hydrocarbon Processing and Handling Installation. The control room could either be blast-resistant or non-blast resistant as applicable to the type of installation.

3.6 EMERGENCY

Emergency means a situation or scenario which has the potential to cause serious danger to person, the environment, or damage to property and which tends to cause disruption inside or outside the premises and may require the help of outside resources.

CLASSIFICATION OF EMERGENCIES: -

Emergencies can be categorized into three broad levels based on seriousness and response requirements, namely: -

(a) Level - I: This is an emergency or an incident which

(i) can be effectively and safely managed, and contained within the site, location, or installation by the available resources.

(ii) has no impact outside the site, location, or installation.

(b) Level - II: This is an emergency or an incident which

(i) cannot be effectively and safely managed or contained at the location or installation by available resource and additional support is alerted or required.

(ii) is having or has the potential to have an effect beyond the site, location, or installation and where external support of mutual aid partner may be involved.

(iii) is likely to be a danger to life, the environment, industrial assets, or reputation.

(c) Level - III: This is an emergency or an incident with off-site impact which could be catastrophic and is likely to affect the population, property, and environment inside and outside the installation, and management and control is done by district administration. Although the Level III emergency falls under the purview of District Authority but till they step in, it should be the responsibility of the unit to manage the emergency.

Note: Level - I and Level - II shall normally be grouped as Onsite emergency and Level - III as an Off-site emergency.

3.7 FLASH POINT

The minimum temperature at which the liquid gives sufficient vapour, when mixed with air, forms an ignitable mixture, and gives a momentary flash on application of a small pilot flame under specified test conditions.

3.8 HVLRM

High-volume long-range monitor having capacity 1000GPM (228 m³/hr) and above.



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3.9 INSTALLATION

For the purpose of this Standard, installation includes all the facilities as listed in 2(i) to 2(iii), which are located in a Refinery, Petrochemicals, and Oil/Gas Processing Plant.

3.10 LPG BOTTLING PLANT

A plant where LPG is stored and filled into cylinders. Receipt and dispatch of LPG by rail, road and pipeline are also considered under this definition.

3.11 OIL/GAS PROCESSING PLANT

A plant where oil and/or natural gas is received, stored and processed to produce Oil/LPG and other petroleum fractions. Gas sweetening, gas processing, desulphurization, etc are normally the process units in any gas processing unit to produce propane, C2/C3, LPG, pentane, naphtha & other higher liquid hydrocarbons.

Upstream facilities are not covered under this definition.

3.12 PETROLEUM DEPOTS, TERMINALS AND OIL & GAS INSTALLATIONS

These include locations where Petroleum crude/Products/LPG are received by tanker, pipelines, tank wagons and tank trucks and stored or blended in bulk and dispatched by tankers, pipelines, tank wagons, tank trucks, portable tanks, or containers. Lube oil installations including lube blending plant / Grease manufacturing plants and can-filling facilities also form part of such installations.

3.13 PETROLEUM REFINERY

A plant where crude oil is received, stored and processed to produce various intermediates and finished products.

3.14 PETROCHEMICAL COMPLEX

Petrochemical complexes are normally petroleum or other hydrocarbon processing industries where chemical, or derivatives are produced from petroleum.

High-Density Polyethylene (HDPE), Linear Low-Density Polyethylene (LLDPE), Butene, PET, PVC, etc are some process blocks in a petrochemical complex.

3.15 SHOULD: 'Should' indicates that the provision is recommended as a good practice.

3.16 SHALL: 'Shall' indicates mandatory requirement.

3.17 MAY: 'May' indicates desirable or good practices which are encouraged to be implemented but not necessary.

3.18 TANK: 'Tank' means a receptacle for petroleum exceeding 1,000 liters in capacity.

3.19 WATER SPRINKLER SYSTEM

Water sprinkler system is a fixed pipe tailor made system to which sprinklers with fusible bulbs are attached. Each sprinkler riser/system includes a controlling valve and a device for actuating an alarm for



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the operation of the system. The system is usually activated by heat from a fire and discharges water over the fire area automatically.

3.20 WATER SPRAY SYSTEM

An automatic or manually actuated fixed pipe system connected to a water supply and equipped with water spray nozzles designed to provide a specific water discharge and distribution over the protected surfaces or area.

4.0 FIRE PROTECTION PHILOSOPHY

Fire Protection Philosophy is based on Control and Loss Prevention. It considers that a hydrocarbon processing plant carries inherent potential hazard. A fire in one part/section of the plant can endanger other sections of the plant as well. If a fire breaks out, it must be controlled / extinguished as quickly as possible to minimize the loss to life, environment and property and to prevent further spread of fire.

4.1 GENERAL CONSIDERATIONS

The size of process plant, pressure and temperature conditions, size of storage, plant location and terrain determine the basic fire protection requirement. Layout of an installation shall be done in accordance with OISD-STD-118 to ensure adequate firefighting access, means of escape in case of fire and also segregation of facilities so that the adjacent facilities are not endangered during a fire.

Fire protection of building shall comply with the National Building Code and Statutory regulations, as applicable.

The following fire protection facilities shall be provided depending upon the nature of the installation and risk involved.

- Fire Water System
- Foam System
- Clean Agent Fire Protection system
- Carbon Dioxide System
- Dry Chemical Extinguishing System
- Detection and Alarm system
- Communication System
- Portable firefighting equipment
- Mobile firefighting equipment
- First Aid Fire Fighting Equipment.

4.2 DESIGN CRITERIA

The following shall be the basic design criteria for a fire protection system.

- i) Facilities shall be designed on the basis that the city fire water supply is not available close to the installation.
- ii) Fire protection facilities shall be designed to fight two major fires simultaneously anywhere in the installation or a single fire for largest floating roof tank, roof sinking case, whichever requires largest water demand.



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Fire water requirement shall be as per guidelines given in Annexure-I.

- iii) Tank farms, process and non-process area shall be fully covered by Fire water Network system.

4.2.1 FIXED WATER SPRAY SYSTEM ON STORAGE TANKS

- i) Class 'A' Petroleum in above ground tanks (floating or fixed roof) shall have fixed water spray system, having capacity larger than 190 m³ or diameter greater than 6 m.
- ii) Class 'B' Petroleum storage tanks of the following dimensions shall be provided with fixed water spray system, whether insulated or uninsulated.
 - a. Floating roof tanks of diameter larger than 30 m.
 - b. Fixed roof tanks of diameter larger than 20 m.

4.2.2 SEMI-FIXED FOAM SYSTEM FOR STORAGE TANKS

Semi-fixed Foam system shall be provided for the following tanks:

- a. Floating roof tanks storing Class 'A' and Class 'B' petroleum products.
- b. Fixed roof tanks storing Class 'A' and class 'B' petroleum products, of diameter ≥ 9 m.
- c. Fixed roof tanks storing class 'C' petroleum products, of diameter larger than 40 m.

Note: Care should be taken in applying portable foam streams to high-viscosity materials heated above 93.3°C (200°F). Good judgment should be used in applying foam to tanks containing hot oils, burning asphalts, or burning liquids that have a boiling point above the boiling point of water. Although the comparatively low water content of foams can beneficially cool such fuels at a slow rate, it can also cause violent frothing and “slop-over” of the tank's contents.

4.2.3 AUTOMATIC ACTUATED RIM SEAL FOAM PROTECTION SYSTEM FOR EXTERNAL FLOATING ROOF TANKS:

An automatic actuated Rim Seal fire detection and extinguishing system shall be provided with the following requirements on all external floating roof tanks storing Class A Petroleum products

Detection System: Linear heat detection system like hollow metallic tube type or two wired detection system or equivalent linear heat detection system.

Extinguishing Media: Foam based extinguishing media.

The major components of the system such as detection system, electrical / electronic components, foam discharge devices etc. shall be listed or approved by national or international agencies such as BIS, UL and FM etc.

Note:- The use of equivalent or superior performance systems / methods to those already prescribed in the standard, should be considered, provided these equivalent systems / methods are duly approved by a technical committee constituted by OISD, based upon technical documentation, performance record and field demonstration by the user industry.



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4.2.4 AUTOMATIC WATER SPRAY FOR STORAGE AND HANDLING OF LPG /C4 AND LIGHTER ENDS/ HYDROGEN :

Automatic Water Spray shall be provided for the following areas:

- i. LPG/C4 and lighter ends and hydrogen, pressure storage vessels shall be provided with automatic water spray system.
- ii. LPG bottling stations, C4 & Lighter ends - loading or unloading gantries, pumps, and compressors.
- iii. Hydrogen compressors associated with storage facilities.

4.2.5 WATER SPRAY SYSTEM IN PROCESS UNIT

Water spray system shall be provided for following hazardous locations equipment in process unit areas.

- a. Un-insulated vessels having maximum design liquid holdup capacity larger than 50 m³ containing class A or B flammable liquid in process unit along with offsite area.
 - b. Vessels inaccessible to fire tender/ mobile equipment, fire hydrants/Monitors.
 - c. Pumps handling petroleum products class 'A' under pipe racks.
 - d. Automatic water spray system for pumps handling LPG or hydrocarbons above auto-ignition temperature.
 - e. Air fin coolers in hydrocarbon service located above pipe racks / elevated location.
 - f. Process compressors of C4 or lighter and Hydrogen.
 - g. Water spray rings for columns of height more than 45 M (above grade level) shall be provided.
- As an alternate, remote operated elevated monitors (at least two, positioned diagonally opposite sides) shall be installed at suitable elevation or high-capacity trolley mounted monitors shall be available, such that they should cover topmost potential source of leakage of column.

Note:- In case there is variance with the recommendations of licensor, then more stringent shall be prevailing.

4.2.6 WATER SPRAY FOR ELECTRICAL INSTALLATION

Water spray requirement with mode of operation to be considered in line with provisions of OISD-STD-173.

4.2.7 WATER SPRINKLER SYSTEM

A water sprinkler system should be provided in polymer extruder buildings, pelletizing building, polymer bagging & its warehouse of a petrochemical complex excluding water reactive/sensitive material storage area. Car parking in the basement of any refinery or petrochemical complex should also be protected with a sprinkler system. For application density & other design aspects of any fire sprinkler system, NFPA-13/IS 15105 may be referred.

4.2.8 WATER CURTAIN SYSTEM FOR NAPHTHA & GAS CRACKER FURNACES

The water curtain shall be provided in cases where hydrocarbon gases are being handled in the vicinity of the cracker furnace in the process unit. The water curtain shall be actuated using Hydrocarbon (HC) detectors placed in the vicinity of the cracker furnace. The water curtain shall be designed as per NFPA 15



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or equivalent at a design flow rate of 37 lpm/m. The water curtain coverage should extend a minimum of 1 m above the flame box of a furnace.

4.2.9 CLEAN AGENT FIRE SUPPRESSION SYSTEM FOR CONTROL ROOM, INSTRUMENT PANEL ROOM & SATELLITE RACK ROOM (SRR)

Clean Agent based fire suppression system should be provided in Control Room, including Satellite Rack Room (SRR), IS data centre and Engineering Room.

Selection of Clean Agent and design of Fire protection system shall follow the Standard on "Clean Agent Extinguishing systems NFPA Standard 2001 (latest edition) or equivalent including its safety guidelines with respect to "Hazards to Personnel", electrical clearance and environmental factors in line with environmental considerations of Kyoto & Montreal protocols and latest MoEFCC guidelines.

Clean Agent System along with all equipment, software for hydraulic calculations and corresponding design concentration of clean agent shall be listed or approved by national or international standards such as BIS/UL/FM/VdS/LPCB etc. Cylinder along with cylinder valve assembly shall have regulatory approval."

4.2.10 LOADING / UNLOADING GANTRY

Class A&B petroleum products or C4 and Lighter ends loading/unloading Tank Truck & Tank Wagon Gantries shall be provided with water spray and/or foam system.

In case automatic fixed water spray system is provided in TW gantry, the gantry should be divided into suitable number of segments (each segment having maximum length of 15 m & width of 12 m) and three largest segments operating at a time shall be considered as single risk for calculating the water requirement.

Accordingly, a provision shall be made to actuate the water spray system from a safe approachable central location i.e. affected zone and adjoining zones.

4.2.11 DRY CHEMICAL FIRE SUPPRESSION SYSTEM

Water reactive material (e.g. TEAL) handling areas including pumps shall be protected with an automatic DCP flooding system. Suitable DCP/metal firefighting DCP shall be used in DCP flooding system for TEAL area. DCP flooding system shall be designed as per NFPA-17.

4.2.12 FIRE PROTECTION FOR COAL HANDLING AND CONVEYOR SYSTEM

Refinery/Petchem facilities where Coal based Captive Power plant available for power and steam generation.

Following Fire protection system shall be installed for the protection of facilities:

- Fire Hydrants, wet risers, and Hose reels
- Fire water Monitors – For structures having height more than 15 m.
- Automatic High Velocity Water spray system for Boiler burner fronts
- Manual medium velocity water spray system with proportioner to mix water with suitable UL approved wetting agent for Coal silos.



- e. Water sprinkler system for open type coke/coal conveyor.
- f. Suitable fire detection system.

4.2.13 FIRE PROTECTION FOR STORAGE / HANDLING OF FLAMMABLE CRYOGENIC LIQUIDS

Cryogenic liquids are liquefied gases that are maintained at very low temperatures (typically less than -150°C). These liquids are gases at ambient temperatures and pressures. Since these liquids are extremely cold and can expand to very large volumes of gases, they are a safety hazard.

- i) Fire measures required for the same are as below.
 - ii) Portable Fire Extinguisher as per Section 12.1
 - iii) Suitable type of PPEs
 - iv) Fire water network / Hydrants & Monitors: as per section 5.6.1 & 5.6.2
 - v) Foam system: Portable / fixed High expansion foam generator shall be provided for protection of:
 - DWST (double walled storage tank) dykes for handling spills.
 - Underground (Pit) liquid transfer pumps.

5.0 FIRE WATER SYSTEM

Water is a universal fire extinguishing agent and meets the majority of ideal properties required for fire extinguishment.

The ideal fire extinguishing agent should have the following characteristics: -

- a. It must be non-flammable
- b. It should have a high heat of vaporization
- c. It should be mobile liquid at normal ambient temperature.
- d. It should be non-toxic and should not decompose to form toxic products.

Thus, water is the best available option and can be used for fire extinguishment, fire control, cooling of equipment and protection of equipment as well as protection of personnel from heat radiation. Water should be used such as straight jet, water fog, water mist, water curtain, water spray, deluge/ sprinkler, for foam making etc.

The firewater system shall comprise fire water storage, fire water pumps and distribution piping network along with hydrants and monitors, as the main components.

5.1 FIREWATER FLOW RATE

The fire water system in an installation shall be designed to meet the fire water flow requirement for fighting two major fire simultaneously or single fire for largest floating roof tank roof sinking case, whichever requiring largest water demand.

Two of the largest flow rates calculated for different sections as shown below shall be added and that shall be taken as design flow rate. An example for calculating the design of major fire water flow rate is given in Annexure-I.

- i) Fire Water flow rate for tank farm shall be aggregated of the following:
 - a) Water flow calculated for cooling a tank-on-fire at a rate of 3 lpm/m² of tank shell area.
 - b) Water flow calculated for all other tanks situated in the same dyke area a rate of 3 lpm/m² of tank shell area



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- c) Water flow required for applying foam into a single largest cone roof or floating roof tank (Rim seal area) burning surface area of oil, by way of fixed/semifixed foam system, where provided or by use of foam monitors. (Refer section 6.8 for foam rates).
- d) Fire water flow rate for supplementary stream, shall be based on using 4 single hydrant outlets and 1 HVLRM (1000 GPM or 228 m³/hr) simultaneously. Capacity of each hydrant outlet (hand line) as 36 m³/hr and of each monitor 228 m³/hr shall be considered at a pressure of 7 kg/cm².
- ii) Fire water flow rate for LPG sphere storage area shall be aggregate of the following:
 - a) Water flow calculated for cooling LPG sphere on fire at a rate of 10.2 lpm/ m² of sphere surface area.
 - b) Water flow calculated for all other spheres falling within a radius of (R+30) meter from center of the sphere on fire at the rate of 10.2 lpm/ m² of surface area.
 - c) Water flow for supplementary stream shall be considered as 372 m³/hr as indicated under annexure(I).
- iii) Water flow rate requirements for firefighting in other major areas shall be calculated based on criteria in terms of lpm/ m² given in section 5.8.

5.2 HEADER PRESSURE

The fire water system shall be designed for a minimum pressure of 7.0 kg/cm² at the hydraulically remotest point of application at grade level and minimum residual pressure of 5.25 kg/cm² at landing valve at elevated structure on designed flow rate at that point.

The fire water network shall be kept pressurized at minimum 7.0 kg/cm² at all the time.

5.3 STORAGE & MAKE-UP WATER

5.3.1 FIREWATER STORAGE

Water for firefighting shall be stored in any easily accessible surface or underground lined reservoir or above ground tanks of steel, concrete, or masonry. The firewater storage shall be located in line with OISD-STD-118.

The effective capacity of the reservoir above the level of pump suction point shall be a minimum of 4 hours aggregate pumping capacity of main pumps (excluding standby pumps). Where the rate of makeup water supply is 50% or more, this storage capacity can be reduced to 3 hours aggregate pumping capacity of main pumps.

The storage reservoir shall be minimum of two equal interconnected compartments to facilitate cleaning and repairs. In the case of tanks there shall be a minimum of two tanks, each having 50 % of the required capacity. During maintenance of tank alternate arrangement of fire water shall be ensured to meet the minimum water requirements.

Large natural reservoirs having water capacity exceeding 10 times the aggregate fire water requirement may be left unlined.

Fire water supply shall be from fresh water source such as river, tube well or lake. Where fresh water source is not easily available, fire water supply can be sea water or other acceptable source like treated effluent



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water. In case sea water or treated effluent water is used for firefighting, the material of the pipe shall be suitably selected.

5.3.2 MAKE UP WATER

Suitable provisions shall be kept for makeup firewater during firefighting time in addition to fire water storage. Provision should be made to divert water from various sources like ETP, Process Cooling Water, river, ponds etc. to the firewater system.

Cautionary– “Water sources should be of high quality and free of water treatment chemicals that will be detrimental to firefighting foam generation. Review and testing before a time of need is prudent to determine whether cooling water additive treatments include antifoaming agents that might affect the quality of firefighting foams.”

5.4 FIREWATER PUMPS

Firewater pumps shall be used exclusively for fire fighting purposes.

5.4.1 TYPE OF PUMPS

Fire water pumps shall be of the following type:

- i) Electric motor driven centrifugal pumps
- ii) Diesel engine driven centrifugal pumps

The pumps shall be horizontal centrifugal type or vertical turbine submersible pumps.

Each pump shall be capable of discharging 150% of its rated capacity at a minimum of 65% of the rated head. The shut-off head shall not exceed 120% of rated head, for horizontal pumps and 140% in case of vertical turbine type pumps.

The number of diesel driven pumps shall be a minimum of 50% of the total number of pumps (inclusive of standby pumps). A minimum of 50% of the total flow requirement should be available through diesel driven pumps all the time.

5.4.2 CAPACITY OF MAIN PUMPS

The capacity and number of main fire water pumps shall be fixed based on design fire water flow rate, worked out on the basis of design criteria as per section 5.1. The capacity of each pump shall not be less than 400 m³/hr. All pumps shall be identical with respect to capacity and head characteristics.

5.4.3 STANDBY PUMPS

- i) In case total number working pumps are up to 2, standby pumps shall be at least one.
- ii) In case number of working pumps are 3 to 4, the number of standby pumps shall be at least 2. For more than 4 working pumps, the number of standby pumps shall be at least 3.
- iii) In cases where two sets of firewater storage and pumps are provided, the number of pumps at each location shall be according to hydraulic analysis of piping network.

5.4.4 JOCKEY PUMPS



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The fire water network shall be kept pressurized at minimum 7.0 kg/cm² by jockey pumps. 2 Jockey pumps (1 working plus 1 standby) shall be provided. The capacity of the pump shall be sufficient to maintain system pressure in the event of leakages from valves etc.

The capacity of jockey pumps shall be minimum 1% of the design fire water rate. Its head shall be higher than the main fire water pumps. Auto cut-in / cut-off facility should be provided for jockey pumps.

5.4.5 POWER SUPPLY FOR FIRE WATER PUMPS

- i) The power supply to the electric motor driven pumps should be distributed from a switchboard having two separate incomers such that failure of any one incomer source does not disrupt the power supply to all the electric motor driven pumps.
- ii) A direct feeder dedicated only to fire water pumps shall be laid from the sub-station to ensure a reliable power supply. The cables for firewater pumps should be laid in a manner that all the cables are not susceptible to damage at all at once.
- iii) The diesel engines shall be quick starting type with the help of push buttons located near the pumps, or at remote location.
- iv) Each diesel engine shall have an independent fuel tank (Preferably outside the pump house) adequately sized for 6 hours of continuous running of the pump.
- v) Main fire water pumps shall start automatically and sequentially with pressure switches/PLC on fire water mains. The system shall ensure auto start of the standby pump in case a pump in sequence fails to start.

5.4.6 LOCATION OF PUMPS

The location and inter-distances for firewater pump house shall follow OISD-STD-118.

5.5 DISTRIBUTION NETWORK

5.5.1 LOOPING & MAINTAINABILITY

The firewater network shall be laid in closed loops as far as possible to ensure multi-directional flow in the system in accordance with NFPA 24. Isolation valves shall be provided in the network to enable isolation of any section of the network without affecting the flow in the rest. The isolation valves shall be located near the loop junctions. Additional valves shall be provided in the segments where the length of the segment exceeds 300 M.

For ease of maintenance, Firewater pumps should be segregated in two groups by providing an isolation valve on common discharge header of pumps.

Flushing connections with isolation valves should be provided at suitable locations in the firewater ring main.

Pressure gauge/Pressure transmitter should be provided at strategic locations (like farthest point) to monitor the header pressure.

For branch piping, an isolation valve shall be provided at the take-off point.

Permanent connection shall not be taken from fire water line / system for purposes other than fire protection/ fire prevention.



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5.5.2 CRITERIA FOR ABOVE / UNDERGROUND NETWORK

The firewater network piping should be laid above ground at a height of at least 300 mm above finished ground level where piping and its components will survive the design accidental loads. Fire water mains shall not pass through buildings or dyke areas, foundations of equipment or structures. However, the fire water network piping shall be laid below ground level at the following places:

- i) Road crossings.
- ii) Places where the above ground piping is likely to cause obstruction to operation and vehicle movement and get damaged mechanically.
- iii) Where frost condition warrants, the ring main system shall be laid underground beneath the frost layer.
- iv) Pipes made of composite material shall be laid underground only.

5.5.3 PROTECTION FOR UNDERGROUND PIPELINES

Where the pipes are laid underground the following protections shall be provided:

- i) The main shall have at least one-meter earth cushion in open ground and minimum 1.2-meter earth cushion under the roads. In the case of crane movement areas, pipes should be protected with concrete/steel encasement.
- ii) The mains shall be provided with protection against soil corrosion by suitable coating/wrapping. Where pipes are coming out/going in from/into the ground, the underground anticorrosion coating on the pipe will continue for a length of at least 300 mm above ground.
- iii) Pipe supports under the pipeline shall be suitable for soil conditions.
- iv) In concrete paved areas, removable paver block should be provided over the buried fire water lines for ease of maintenance without cutting of pavement or fire water line should be laid in concrete trenches which is filled with sand and covered with suitable strength slabs for ease of maintenance without cutting of pavement.

5.5.4 PROTECTION FOR ABOVE GROUND PIPELINES

Where the pipes are laid above ground, the following protection shall be provided:

- i) The firewater mains shall be laid on independent sleepers by the side of road. These shall not be laid along with process piping on common sleepers.
- ii) The mains shall be supported at regular intervals adequately as per design.
- iii) The system for above ground portion shall be analyzed for flexibility against thermal expansion and necessary expansion loops shall be provided wherever called for.

5.5.5 HYDRAULIC ANALYSIS & SIZING OF FIREWATER NETWORK

- i) The hydraulic analysis of the network shall be done in accordance with NFPA 14, NFPA 15 and NFPA 24. Also, whenever fire water demand increases due to addition of plant & facilities, extension of network, fresh hydraulic analysis shall be carried out.
- ii) Fire water distribution ring main shall be sized for 120% of the design water rate. Design flow rates shall be distributed at nodal points to give the most realistic way of water requirements in an emergency. Several combinations of flow requirements shall be assumed for the design



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of the network. For large volume of water requirement for floating roof tank (Annexure-V), the network around tank farm shall be suitably designed.

5.5.6 FIRE HYDRANTS

Fire water hydrants shall be provided on the fire water network (Ref. Section 5.6 for details). Each of these connections shall be provided with independent isolation valves. Refer item 5.6.1 for further details on hydrants.

5.5.7 FIXED WATER MONITORS

Fixed water monitors shall be provided on the firewater network (Ref. Section 5.6 for details). Each of these connections shall be provided with independent isolation valves.

5.5.8 LAYOUT

- i) Fire water mains shall not pass through buildings or dyke areas.
- ii) Hydrants / monitors shall not be located inside the dyke area.

5.6 HYDRANTS & MONITORS - DETAILS

5.6.1 HYDRANTS

- i) Hydrants shall be located considering the fire hazards at different sections of the premises to be protected and to give the most effective service. At least one hydrant post shall be provided for every 30 M of external wall measurement or perimeter of unit battery limit in case of hazardous areas. Hydrants protecting utilities and non-plant buildings can be spaced at 45 M intervals. The horizontal range and coverage of hydrants with hose connections shall not be considered more than 45 M.
 - ii) The hydrants shall be located at a minimum distance of 15 M from the periphery of storage tank or hazardous equipment under protection. For process plant's location of hydrants shall be decided based on coverage of all areas. In the case of buildings, this distance shall not be less than 5 M and more than 15 M from the face of the building.
 - iii) Hydrants / Monitors shall be located along roadside berms for easy accessibility as far as possible. Hydrants/ Monitors shall not be positioned on dyke wall.
 - iv) Double headed (minimum 4" bore size standpost with two independent Type-A landing valves) shall be used. All hydrant outlets shall be situated at a workable height of about 1.2 meters above ground level.
 - v) Hydrants / Monitors shall be located with branch connections and not directly over main header for easy accessibility and to avoid corrosion of main header due to dripping from hydrants/accumulation of water.

5.6.2 MONITORS

- i) Monitors shall be located at strategic locations for protection of columns, heaters, gasifiers, etc., and where it is not possible to approach the higher levels. A minimum of 2 monitors shall be provided for the protection of each such area. Water monitors for protection of heaters shall be installed so that the heater can be isolated from the remainder of the plant in an emergency.



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ii) Monitors shall be located to direct water on the object as well as to provide water shield to firemen approaching a fire. The monitors, including HVLRM shall be installed at a minimum of 15 m away from the equipment or facilities to be protected. However, the distance of monitors shall not exceed the performance range of the monitor from the hazard to be protected.

iii) Field adjustable variable flow monitors shall be installed at critical locations. These shall be in compliance to the national or international standards such as BIS, UL and FM standard etc.

The requirement of monitors shall be established based on hazard involved and layout considerations.

The location of monitor is to be planned in such a way that the very purpose of these monitors are served, and the throw of the monitors is safely delivered at the aimed object.

Remote operated Monitors can be placed at elevated platform to increase their vertical reach to protect column. Operating valve for such monitors should be kept at grade level for its operation during the requirement.

Hydrant/Monitors should be painted in line with OISD-GDN-115.

5.6.3 Dry/Wet Risers with hydrants should be provided on each floor of technological structures. It shall be provided on both sides if technical structure length is more than 30m and stair case access provided from both ends

5.6.4 FIXED REMOTE / MANUAL OPERATED, MOBILE HIGH VOLUME LONG RANGE WATER OR WATER CUM FOAM MONITORS

Remote / Manual operated high-volume long-range water or water cum foam monitors, variable flow type nozzle (capacity 1000 GPM or 228 m³/hr and above) shall be provided in

i) Inaccessible areas such as column, reactor, compressor house etc. in critical units like CCRU, DHDS, HCU, Hydrogen, FCCU, DCU, CDU etc.

ii) Critical equipment at higher elevation (above 45m)

iii) Tank farm area

These monitors can be fixed type or mobile trailer mounted for ease of operation and efficacy of foam application.

Below criteria should be followed for provision of water cum foam monitors in tank farm area.

a) The number and capacity of monitors shall be provided in such a way that cumulative foam application rate from the designated monitors meet the application rate of 9.75 lpm/m² for total burning surface of the affected tank to extinguish the full surface tank fire. Additional monitors shall be provided in such a way that each tank is in the coverage area of at least two monitors. Due consideration has been given to compensate for the losses (50%) due to wind and other factors.

b) Monitors should be located to achieve the foam discharge preferably close to tank center from upwind direction of tank on fire.

c) A suitable connection for water supply to portable trailer mounted HVLR should be made available.

The basic water cum foam monitors shall be listed or approved by national or international agencies such as BIS, UL and FM etc. In case such certification is not available (applicable only for high-volume mobile



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trailer-mounted monitors), the equipment shall be certified by an approved Third-Party Inspection (TPI) agency recognized by the entity. The electrical or hydraulic remote-control mechanism shall be conforming to Hazardous Area Classification.

5.6.5 HOSE BOXES OR STATIONS

Provision of hose boxes or stations should be given based on the risk assessment for housing hoses and nozzles.

5.6.6 WATER CUM FOAM MONITORS FOR GANTRY AREA

Tank Wagon & Tank Lorry loading/ unloading gantry area shall be provided with alternate water cum foam monitors having multipurpose combination nozzles for jet spray & fog arrangement and fire hydrants located at a spacing of 30m on either side of the gantry. These monitors shall be complying to national or international standards such as BIS, UL and FM etc.

(This is in addition to the water spray requirement given in 4.2.10)

5.7 MATERIAL SPECIFICATIONS

All the materials required for a firewater system using fresh water shall be of approved type as indicated below and in line with OISD-GDN-115.

- i) **Pipes:** Carbon Steel as per IS: 3589/ IS: 1239/IS:3183 or Composite materials as per API 15LR/API 15 HR or its equivalent shall be used.

In case saline water/ treated effluent water is used, the fire water main of steel pipes shall be internally cement mortar lined or glass reinforced epoxy coated or made of pipe material suitable for the quality of water. The material selection for the fire main line shall consider the quality or impurities in the water. Alternatively, pipes made of composite materials shall be used. Cast iron pipes shall not be used for fire water services.

- ii) **Isolation Valves:** Cast Steel valves *shall be used* in *all areas* including unit areas, *offsite* and fire water pump stations.

Isolation valves having open/close indication shall be Gate valves. More than 12" should be provided with gear mechanism.

5.8 FIXED WATER SPRAY SYSTEM

5.8.1 GENERAL

It is a fixed pipe system connected to a reliable source of water supply and equipped with water spray nozzles for specific water discharge and distribution over the surface of area to be protected. The piping system is connected to the Fire water network water supply through an automatically or manually actuated valve which initiates the flow of water.

- i) A fixed water spray system should be provided in high hazard areas where immediate application of water is required (refer 4.2.1, 4.2.5, 4.2.6, 4.2.10 above).
- ii) Water supply patterns and their densities shall be selected according to need. The fire water spray system for exposure protection shall be designed to operate before the possible failures of any containers of flammable liquids or gases due to temperature rise. The system shall, therefore, be designed to discharge effective water spray within the shortest possible time.



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5.8.2 WATER SPRAY APPLICATION RATES

The following water spray application rates are recommended for general guidance. These rates should be reviewed on a case-to-case basis and increased, if required. While calculating the water rates for spray application for cases other than tanks/vessels, the area should be divided into suitable segments so that maximum water requirement for spray application should generally not exceed 1200 m³/hr.

Application Area	Water application rate
Atmospheric storage	3 lpm/m ² of tank shell area for tank on fire
Tanks	3 lpm/m ² of tank shell area for exposure protection for tanks located within in the same dyke area.
Pressure Storage Vessels	10.2 lpm/m ² of shell area
Process Unit Area	
- Pumps (Volatile product service located under Pipe rack), Compressors	20.4 lpm/m ²
- Columns, other Extremely hazardous area	10.2 lpm/m ²
- LPG pump house	20.4 lpm/m ²
- LPG Tank Truck & Tank Wagon loading/ Unloading gantries	10.2 lpm/m ²
- LPG Bottling plants:	
- Carousel machine	10.2 lpm/m ²
- Filled cylinder storage	10.2 lpm/m ²
- Empty cylinder storage	10.2 lpm/m ²
- LPG cylinder cold repair. Shed	10.2 lpm/m ²
- Oil Tank Truck & Tank Wagon loading/unloading gantries	10.2 lpm/m ²
- Cable Trays	10.2 lpm/m ²
- Transformers (Above 10MVA)	10.2 lpm/m ²

5.9 FIXED WATER SPRINKLER SYSTEM

- i) Sprinkler systems are used for fire extinguishment when the hazards are located inside buildings. Some of the examples being: Building/sheds storing combustible and flammable materials.
- ii) The water for the sprinkler system shall be tapped from the plant fire hydrant system, the design of which should include the flow requirement of the largest sprinkler installation.



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- iii) Minimum Design density and assumed area of operation of fixed water sprinkler system shall be in accordance with hazard area occupancies classification given in IS 15105 or NFPA 13.
- iv) The design flow for sprinkler installation would depend on the type of hazard and height of piled storage. The design water flow shall be restricted to a minimum of 100 m³/hr and to a maximum 200 m³/hr. The design flow rate for other areas shall be taken as 10.2 lpm/ m² of the area protected by sprinkler installation, subject to a minimum of 150 m³/hr. and a maximum of 400 m³/ hr. Higher water application rates should be used if called for depending on risk involved.

6.0 FOAM SYSTEM

For combating large hydrocarbon fire particularly in a contained area like storage tank, foam has proved useful for its inherent blanketing ability, heat resistance and security against burn-back. Efficient and effective foam delivery system is a vital tool for its usefulness in controlling fire. The process of adding or injecting the foam to water is called proportioning. The mixture of water and foam compound (foam solution) is then mixed with air in a foam maker for onward transmission to burning surface.

6.1 TYPES OF FOAM

For detailed description of Fire Fighting Foam Compounds, please refer Annexure-VIII.

6.2 FOAM CONVEYING SYSTEMS

The system consists of an adequate water supply, supply of foam concentrate, suitable proportioning equipment, a proper piping system, foam makers and discharge devices designed to adequately distribute the foam over the hazard.

Conventional systems are of the open outlet type in which foam discharges from all outlets at the same time, covering the entire hazard within the confines of the system. It shall be ensured that uninterrupted foam concentrate supply from source to foam equipment is available.

There are three types of systems:

- i) Fixed
- ii) Semifixed
- iii) Mobile

6.2.1 FIXED FOAM SYSTEM

Fixed foam conveying system comprises of fixed piping for water supply at adequate pressure, foam concentrate tank, inductor or any other suitable proportioning equipment for drawing foam concentrate and making foam solution, fixed piping system for onward conveying to foam makers for making foam, vapour seal box, and foam pourer

6.2.2 SEMIFIXED FOAM SYSTEM

Semi-fixed foam system gets supply of foam solution through the mobile foam tender. A fixed piping system connected to foam makers cum vapour seal box in case of cone roof tanks and foam maker and foam pourers in the case of floating roof tanks conveys foam to the surface of tank.

6.2.3 MOBILE FOAM SYSTEM



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Mobile system includes foam producing unit mounted on wheels which can be self-propelled or towed by a vehicle. These units supply foam through monitors/foam towers to the burning surface.

6.2.4 OTHER SYSTEMS:

Automatic Actuated Foam Flooding system:

In this system all the components and ingredients including premix solution are contained within the system. Such systems usually have a premix solution supply tank pressurized by air or inert gas. The automatic sensing of fire releases this pressure and places the system into operation. The minimum requirement in design for the automatic actuated foam flooding system is given in Annexure VII.

6.3 FLOATING ROOF TANK PROTECTION USING FOAM

6.3.1 PROTECTION USING SEMI-FIXED FOAM SYSTEM:

For floating roof tank, foam shall be poured at the foam dam to blanket the roof's rim seal. Features of foam system for floating roof tank protection shall be as follows:

- i) System shall be designed to create foam blanket on the burning surface in a reasonably short period.
- ii) Foam shall be applied to the burning hazard continuously at a rate high enough to overcome the destructive effects of radiant heat.
- iii) Foam maker's/foam pourers shall be located not more than 24 m apart on the shell perimeter based on 600 mm foam dam height. The height of foam dam shall be at least 51 mm above the top of metallic secondary seal.
- iv) A minimum of two foam pourers shall be provided.

6.3.2 PROTECTION USING AUTOMATIC ACTUATED FOAM FLOODING SYSTEM:

Provision of an automatic rim-seal protection system of foam flooding type shall be in line with the details mentioned at 4.2.3 and at Annexure-VII.

6.4 FIXED ROOF TANK PROTECTION USING FOAM

Foam conveying system shall have same features as of floating roof tank excepting that a vapour seal chamber is required before the foam discharge outlet.

Features of the foam system for fixed roof protection for class A & B shall be as follows:

- i) The vapour seal chamber shall be provided with an effective and durable seal, fragile under low pressure (at an inlet pressure of not less than 10 psi (69 kPa) and not greater than 25 psi (172 kPa), to prevent entrance of vapour into the foam conveying piping system.
- ii) Where two or more vapour seal chambers are required, these shall be equally spaced at the periphery of the tank and each discharge outlet shall be sized to deliver foam at approximately the same rate.
- iii) Tanks should be provided with foam discharge outlets/ vapour seal chambers as indicated below:

Tank Diameter in m.	Minimum number of foam discharge outlet
≥ 9 upto 20	2
>20 upto 25	3



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>25 upto 30	4
>30 upto 35	5
>35 upto 40	6
>40 upto 45	8
>45 upto 50	10
Plus 1 outlet for each additional 465 m ² of area	

The estimation of number of foam discharge outlet is based on pourer capacity of 1000 lpm at a pressure of 7 kg/cm² upstream of eductor. This can be suitably adjusted for different vapour seal chamber capacity in accordance with para 6.3.1 (iii).

6.5 FLOATING CUM FIXED ROOF TANK PROTECTION USING FOAM

Protection facilities shall be provided as required for fixed roof tank.

6.6 FULL SURFACE STORAGE TANK FIREFIGHTING & LOGISTICS

Large Diameter Full Surface Tank Fire Fighting:

Full surface Tank Fire Scenario:

Part A: Tanks diameter up to 45 meter

It is generally experienced that full surface fire in smaller diameter petroleum storage tank up to 45-meter diameter are relatively easy to extinguish with reliable resources and competent manpower.

This shall be achieved by designing dyke foam monitors of adequate capacity / foam tender roof monitors. Un-interrupted foam supply shall be ensured by any reliable means for a continuous period of 65 minutes.

Part B: Tanks diameter larger than 45 meters

Large diameter full surface tank fire is a challenge and requires significant preparation in terms of resources and well proven strategy.

Successful extinguishment of a large diameter petroleum storage tank full surface fire begins with pre-fire planning and therefore most of the following information shall be considered. Entity to prepare the pre-fire plan for largest diameter tank at site and list out the resources required. The priority shall be for rescue and evacuation of personnel before attempting extinguishment.

The manpower requirement to tackle a major tank fire may vary depending upon the type, location and nature of the fire, method of extinguishment and the availability of trained personnel. This shall be determined while developing the pre-fire plans.

Resources Requirement:

Below listed major resources shall be made available at site as a design requirement and periodically these resources should be tested and maintained in ready to use condition. The critical aspect of successful tank fire extinguishment is a stable uninterrupted foam delivery in required quantity through aerial (over the top) application. Foam discharge through large volume monitors should reach close to



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tank center for effective foam blanketing over burning surface. Many successful full surface tank fire fighting are achieved by throwing foam through 2-3 large volume foam monitors from upwind side of the burning tank. Multiple direction foam discharge over a burning tank may not give a satisfactory result.

A typical list of major Trolley /portable / mobile equipment for full surface tank fire is given below.

- a. High-Volume Long-Range foam monitors
- b. Water monitors
- c. Foam pumping/ proportioning appliances
- d. Foam concentrate tankers or containers (Tote containers)
- e. Fire hose including large diameter/ capacity hose and Water supplies

a. High Volume Long Range (HVL) Foam Monitors

The foam monitor capacity and range are important factors for fire extinguishment. Foam streams have to be such that the bulk of the discharge reaches the tank liquid surface. It is recognized that the best method of application is to project foam with the wind behind the stream and not against. However, there have been, and will be situations where cross winds or a variable breeze causes destructions in stream range and these needs to be considered. Therefore, it is important to use large capacity trailer mobile foam monitors for tank fire extinguishment, which give the versatility for placement considering the prevailing wind direction so that the bulk of the output reaches the tank liquid surface without affecting the stream range. Number and capacity of large volume foam monitors should be decided based on Pre-fire plan for largest diameter tank fire. Foam monitors are available ranging from 1000 GPM (228 m³/hr) to 12000GPM (2736 m³/hr).

b. Water Monitors

Water monitors are necessary for cooling adjacent tanks affected by radiant heat or direct flame impingement. Water cooling of the tank on fire is not normally necessary except to assist foam blanket sealing against a hot tank wall. Cooling may be attempted if the water spray system is intact and operable for uniform cooling.

c. Foam proportioning appliances

An uninterrupted foam supply with required induction rate matching with foam monitors discharge for the total duration of foam application is necessary for extinguishment. It is very essential that the selected foam proportioning system is suitable for the logistical challenges involved in bulk foam concentrate handling. It is possible to supply foam to the foam monitors either using foam pumps, or foam fire tenders or through foam pick up of a monitor.

d. Foam concentrate and supply considerations

Availability of required quantity of foam concentrate at emergency site shall be ensured before commencing foam application. Once foam application commences, it shall be sustained for a minimum period of 65 minutes @ 9.75 lpm/m². It is recommended that bulk storage is provided in foam nurer trucks, elevated bulk storage tanks or 1000-liter Tote containers on mobile trailers / trucks. Conventionally 3% foam concentrates are used for tank fire fighting, however 1% foam concentrate



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can be considered for better logistics management. It is highly recommended to keep the total quantity of foam as a separate stock to be mobilized only during tank fire scenario.

It is experienced that foam losses during actual firefighting is high around 50-60 % during heavy wind or adverse weather condition or flame draft from the burning tank. Safety allowance of additional 50% has been included in the above application rate.

Foam compatibility is an important factor. Mixing foam concentrates of different types is not recommended and can destroy foam blanket over the burning surface. Moreover, foam concentrate should be compatible with associated resources being used for firefighting.

Suggested options for Foam concentrate supply-

- Foam Nurser / Foam Transport Vehicle / Foam Tender
- Foam Tote on Trailer with common discharge manifold

e. Fire Hose and Water supply

Wherever the full surface firefighting strategy is adopted with mobile trailer mounted high-capacity water cum foam monitors, large capacity hoses for feeding water to the monitor should be selected to avoid losses / pressure drop during actual operation. The sizes of large capacity hoses vary from 4" to 12". Conventional 63mm fire hoses may be used for portable water monitors for cooling and foam proportioning system. Power driven device may be used for quick laying of hoses.

Fire water from available fire water network / large capacity pumps should be used to feed large diameter hoses for monitors. Suitable feeding arrangement should be made available for this water supplies.

Note: For foam calculation refer Annexure-VI.

6.7 DYKE AREA/ SPILLS/ OIL SEPARATOR PROTECTION USING FOAM

Portable HVLRM of suitable range/foam hose streams shall be provided for fighting fires in dyked area and spills. Additionally, Portable/Fixed Medium expansion foam generators shall be provided to arrest vapor cloud formation from spilled volatile hydrocarbons.

6.8 FOAM APPLICATION RATE

The minimum delivery rate for primary protection based on the assumption that all the foam reaches the area being protected shall be as indicated below. In determining total solution flow requirements, potential foam losses from wind and other factors shall be considered.

For cone roof tanks containing liquid hydrocarbons, the foam solution delivery rate shall be at least 5 lpm/m² of liquid surface area of the tank to be protected.

For floating roof tanks containing liquid hydrocarbons, foam solution delivery rate shall be at least 12 lpm/m² of seal area with foam dam height of 600 mm of the tank to be protected. In case of floating roof sinking, the rate considered should be 9.75 lpm/m² of liquid surface areas for 65 minutes. (Refer Annexure-V for sample calculation)

DURATION OF FOAM DISCHARGE



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The equipment shall be capable of providing primary protection at the specified delivery rates for the following minimum period of time.

- i) Tanks containing liquid hydrocarbons - Class 'C' Petroleum - 30 minutes.
- ii) Tanks containing Class 'A' & 'B' Petroleum or liquids heated above their flash points - 65 minutes.
- iii) Where the system's primary purpose is for spill fire protection - 30 minutes.

Note:- It is advised to keep foam compound stock for extended 30 minutes for full surface tank fire scenario.

6.9 FOAM QUANTITY REQUIREMENT

The aggregate quantity of foam solution for a single largest tank fire should be calculated as the sum of foam quantity requirement in clause 6.8 and total indicated below under items (i) & (ii) for a minimum period of 65 minutes. The quantity of foam compound required should be calculated based on 1%, 3% or 6% concentrate. Necessary changes in the equipment shall be made accordingly. (Refer Annexure-VI for sample calculation)

- i) One foam monitor of 1000 GPM (228 m³/hr) foam solution capacity.
- ii) Two hose streams of foam each with a capacity of 900 lpm of foam solution. A typical example showing calculation of foam compound requirement is given at Annexure - II.

6.10 FOAM COMPOUND STORAGE

- a. Foam compound should be stored in containers of suitable capacity, in case of protein, fluoro protein or AFFF or AR-AFFF or FFFP. Alcohol Resistant Foam (AR-AFFF) shall be used for handling methanol/ ethanol fires. Alcohol Resistant Foam (AR-AFFF) can be used for all type of hydrocarbon fire and polar solvent fire effectively. Foam compound can also be stored in overhead storage tank of suitable capacity for quick filling of foam tender / nurser during emergency.

Cautionary: Alcohol Resistant Foam (AR-AFFF) is preferred for Ethanol Blended Petrol (EBP) fire.

- b. Type of foam compound used can be protein or fluoro-protein or AFFF or AR-AFFF or FFFP. Maximum life of foam compound shall be taken as per manufacturer's data.
- c. Foam compound shall be tested periodically in line with OISD-STD-142.
- d. Quantity of foam compound equal to 100% of requirement as calculated in 6.9 should be stored in the Installation. However, for installations having largest tank of diameter as 79m, minimum of 93230 (~31077 in case of 1% foam) liters foam should be stored (refer Annexure VI) or foam sufficient to fight two major fire whichever is more.

Note: Alcohol Resistant Synthetic Fluorine Free Foam (AR-SFFF) are environmentally friendly foam may be used.

7.0 CLEAN AGENT BASED PROTECTION SYSTEM FOR CONTROL ROOM , SRR AND COMPUTER ROOM PROTECTION

Selection of Clean Agent and system design for fire protection shall be follows



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Gas (Clean Agent) based automatic fire detection and extinguishing system should be provided to protect electronics housed in cabinet, electrical equipment, Cables and like other devices. located in satellite rack rooms (SRR) or in control rooms.

Selection of Clean Agent and design of Fire protection system for process control rooms and SRR shall follow the applicable NFPA Codes or Standards including its safety guidelines with respect to “Hazards to Personnel”, electrical clearance and environmental factors in line with environmental considerations of Kyoto & Montreal protocols and latest MoEFCC guidelines.

QUANTITY AND STORAGE OF CLEAN AGENT

Each hazard area to be protected by the protection system shall have an independent system.

The time needed to obtain the gas for replacement to restore the systems shall be considered as a governing factor in determining the reserve supply needed. 100% standby charge of clean agent containers shall be considered for each protected hazard.

Storage containers shall be carefully located so that they are not subjected to mechanical, chemical, or other damage.

All the components of the system shall be capable of withstanding the heat of fire and severe weather conditions.

8.0 BATTERY FIRE

From consumer electronics and electric vehicles to industrial equipment and energy storage systems, lithium-ion batteries can be found everywhere. Due to the materials utilized to manufacture these batteries, the scene of a lithium-ion battery fire can best be compared to a hazmat (hazardous material) incident. Although extinguishing fire is a primary concern, flammability, explosivity and toxicity must be addressed to ensure proper protection and prevent re-ignition.

In a cell or battery, ignition occurs when lithium-ion batteries enter a chemical process called thermal runaway. Cells produce heat and emit explosive and toxic gases, developing a self-propelling loop that leads to thermal runaway propagation. This chain reaction will continue to impact adjacent cells until thermal runaway propagation is halted. Confined spaces such as server rooms, automobiles, aircraft, spacecrafts, shipping containers, energy storage systems and more can pose an increased risk.

During the battery fire, a flammable electrolyte used within cells catches fire. Thus, fire starts inside a cell which makes it difficult to reach. All the conventional firefighting media, including plain water, works at the macro / surface level to remove heat. In the case of battery fire, due to deep seated nature of fire, the effectiveness of conventional firefighting media is very less and thus they are required in humongous quantity to achieve extinguishment.

In many battery fires, one of the byproducts of the oxidation (burning) chemical process is oxygen, thus the fire is fed by creating its own oxygen. Hence, any firefighting media / techniques that works on a principle of oxygen depletion find limited success.

8.1 FIRE PROTECTION

For fire protection of Li ion battery, Encapsulator Agent (EA) based on spherical micelle technology should be used in line with NFPA 18A & NFPA 855 (latest edition). (For details refer Annexure-X)



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9.0 CARBON DIOXIDE SYSTEMS

Fixed CO₂ system shall be provided in Turbo generator enclosure, Gas turbine enclosure etc as per OISD STD 173. A fixed CO₂ system should be designed and installed in accordance with NFPA-12.

Before the CO₂ flooding system is operated, persons in the confined area should be evacuated.

Carbon dioxide is an odourless and colourless inert gas having a proven fire extinguishing property. When applied in a proper manner and in proper quantities on a fire hazard, it extinguishes fire by cutting off the oxygen and creating an inert atmosphere around the hazard. If the inert atmosphere is maintained for a reasonable time the possibility of flash back is also reduced.

Carbon dioxide is best applied to a fire hazard through a fixed system, consisting of CO₂ storage, distribution piping and discharge nozzles.

10.0 DRY CHEMICAL EXTINGUISHING SYSTEM

The extinguishing system comprises of supplying the Dry Chemical agent, manually or automatically, through a distribution system onto or into the protected hazard.

10.1 RECOMMENDED USE

A dry chemical powder extinguishing system should be used as per suitability on following hazards.

- Electrical hazard such as transformers or oil circuit breakers.
- Pyrophoric materials and alkyl metal compounds (Such as TEAL, TEB) etc.
- Combustible solids having burning characteristic like naphthalene or pit which melts while on fire.
- Class 'A', 'B', 'C' & 'D' fires.

Requirement for each item should be finalized while deciding on the design basis.

10.2 SYSTEM DESIGN

Basic requirement of designing the dry chemical extinguishing system is to provide for sufficient quantity and rate of discharge depending upon the hazard.

The system consists of dry chemical powder and expellant gas container assembly of capacity sufficient for given hazard with distribution piping and discharge nozzles. System can be actuated manually or automatically on visual or automatic means of detection. Alarm and indication shall be provided to show that the system has operated, and a personnel response is needed.

Personnel safety shall include training, warning signs, discharge alarm, respiratory protection, and prompt evacuation of personnel.

Following types of systems can be provided to protect a hazard:

- Total flooding system
- Local application system
- Hand hose line system, and
- Pre-engineered system

(Refer NFPA-17 for limitations & precautions for use of dry chemical and for system design)

11.0 MODERNIZATION PLAN FOR FIREFIGHTING EQUIPMENT



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As a part of modernization of firefighting equipment, following may be explored in addition to the OISD-GDN-115 based on its suitability to hazards, long term maintainability, reliability and certification.

a. Turbine Based Water Mist System

Turbine based water mist system is a compact, high flow water/ fog generator. The entire system is of modular design which includes turbine system, fuel tank and water/ foam supply. By injecting water, in the exhaust stream of the gas turbine a fog is produced. Due to the power of turbine, this fog can travel a large distance and can be controlled and influenced by the turbine speed, water nozzle configuration and water pressure. The fog stream can be pointed to the object (fire, tank, gas cloud) with a joystick which hydraulically controls the angle of the turbine, up or down and left or right.

b. Robotic Monitor

A Robotic monitor is a self-driven remote-controlled unit which is capable to travel into areas which are considered unsafe, and hazardous for fire fighters to approach. The unit comprises of a water/ foam monitor, nozzles for producing water mist, thermal/ visual imaging camera, gas detector, cable winch and a dozer blade (optional). The unit moves on heavy duty rubberized tracks enabling it to cross over open drains and trenches.

c. Aerosol Based Fire Suppression System

The aerosol employs a fire extinguishing agent consisting of very fine solid particles and gaseous matter to extinguish fires. The condensed aerosol micro particles and effluent gases are generated by the exothermic reaction; until discharged from the device, the particles remain in vapor state. They are cooled and "condensed" within the device and discharged as solid particles. The condensed aerosol agent can be delivered by means of mechanical operation, electric operation, or combined electro-mechanical operation.

d. Dual Agent Nozzle

The nozzle is designed to throw liquid (foam solution / water) and DCP simultaneously. The liquid forms a hollow jet and the DCP is inside the hollow space. Application wise DCP provides quick knock down of flames and the liquid provides cooling. Also, DCP can be thrown at a longer distance than conventional DCP hand nozzle or monitor.

e. Fire extinguisher of HDPE / Composite material body

This will ensure less weight and claimed to have longer shelf life. This will be more useful in corrosion prone area like offshore.

12.0 FIRST AID FIRE FIGHTING EQUIPMENT

12.1 CRITERIA TO DETERMINE THE QUANTITY NEEDED:

Portable firefighting equipment shall be provided in Refinery/Petrochemical/Process plant as indicated in the following table:

Description	Norms/criteria to determine the quantity needed
Conventional firefighting equipment	
i) Dry chemical powder (DCP) fire extinguishers 9kg capacity: IS:15683/ UL299	While selecting the Extinguisher, due consideration shall be in line with OISD-GDN-115/ IS: 2190 / UL711. Extinguisher to be located in process units, pump houses, pump area, LPG storage area, LPG bottling plant, Oil separator, tank



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	truck/ tank wagon loading areas, substations, Workshops, laboratory, power station buildings etc. The number should be determined based on the maximum traveling distance of 15 M in the above areas. At least one fire extinguisher shall be provided for every 250 m² of hazardous operating area. Fire extinguisher should also be considered at elevated location such as technical structure. There shall be minimum two extinguishers at a location e.g. pump house.
ii) Dry chemical powder fire extinguishers 25/50/75 kg capacity: IS:16018/ UL299 (in addition to 9kg DCP requirement)	The extinguishers, with the selection criteria viz. flow rate, discharge time and throw as mentioned in IS 2190, to be located in operating areas. At least one fire extinguisher should be provided for every 750 m² of hazardous operating area.
A. Pump House (Class A/B) Up to 50 HP Above 50-100 HP Beyond 100 HP	1 No. 9 Kg DCP for 2 pumps. 1 No. 9 Kg DCP for each pump. 2 Nos. of 9 kg or 1 no. of 25 kg DCP for each pump.
B. Pump House (Class C) Up to 50 HP Above 50 HP	1 no. 9Kg DCP for every 4 pumps up to 50 HP. 2 nos. 9 Kg DCP or 1x25 kg DCP for 4 pumps.
C. Tank Wagon loading and unloading gantry/siding	1 No. 9 Kg DCP extinguisher for every 30 m of gantry/siding plus 1 No. 75 Kg DCP extinguisher for each gantry/siding.
iii) CO2 extinguishers 4.5/6.5/9.0/22.5 kg capacity (IS:16018/IS15683/UL154)	To be located in substations, power stations, office building and control room. The number should be determined based on the maximum traveling distance of 15 m. At least one fire extinguisher shall be provided for every 250 m² of hazardous operating area. There shall not be less than 2nos. extinguishers at one designated location e.g. control room.
iv) Class F fire extinguisher	To be located in canteen area.
v) Portable Encapsulator agent extinguisher	To be located in battery rooms storing lithium-ion battery.
vi) Portable clean agent extinguishers	This should be as an alternate to a CO2 extinguisher. To be located in control rooms, computer rooms, laboratories, and office buildings.
vii) Portable water-cum-foam monitor.	Minimum 2 units for Petroleum refinery and 1 unit for Gas Processing Plant
viii) Portable Foam Trolley (Capacity max 200 liter)	As per requirement
ix) Steam lancers (as a part of utility station)	For fighting incipient fire at flange leaking & hot pumps.
x) Rubber hose reel (25mm)	To be located in Process unit battery limits and other process areas for quenching of incipient fire.



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Note : For substations, power station; the number and area for placement of extinguishers should be decided as per “OISD STD-173” requirements.

12.2 OTHER EMERGENCY EQUIPMENT:

Additionally, the following items should also be provided. The number of units required for these shall be decided by local management, on a case-to-case basis.

- i) Thermal Imaging Camera: As an aid to the fireman during firefighting operation to locate the seat of the fire and to facilitate search and rescue operation in smoky area e.g. during cable gallery.
- ii) Personal Protective Equipment required during Fire Fighting like Water gel- based blanket, Fire Proximity Suit, Self-contained breathing apparatus, Airline breathing apparatus, Safety Helmets, Fire Helmets, Stretcher, First Aid box, Rubber hand gloves, canister mask etc.
- iii) Other Equipment like Portable Gas detectors, Explosive meter, Oxygen meter, Hand operated siren, Red/Green Flag, Safe Walk roof top ladder, portable emergency lighting, intrinsically safe portable megaphone, various leak plugging gadgets, oil dispersants and oil adsorbents, lifting bags/jacks (for rescue of trapped workers).

13.0 MOBILE FIRE FIGHTING EQUIPMENT
13.1 FIRE TENDERS:

The exact number of fire tenders shall be higher of the items (a) or (b):

- (a) The quantities fixed up in each case based on two simultaneous major fire taking into consideration the size, location of the plant and statutory requirements.
- (b) The quantities indicated below.
 - i) 3 nos. of foam tenders out of which two are for firefighting, one for spill /standby. The foam tender should have a foam tank capacity of at least 3000 Liter and the pump capacity of minimum 4000 lpm at 8.5 kg/cm².
 - ii) DCP tender(s) having aggregate 4000 kg capacity, with Nitrogen as expellant gas. These are required for fighting LPG/Gas fire. A monitor should have a variable throw of 15/25/40 kg/sec. The throw of the monitor shall be 40 to 50 M for the DCP charge.

Minimum 1 number of Foam Nurser with foam compound tank of minimum 8000 litre capacity, with suitable pump for foam transfer. Entity shall ensure the continuous availability of foam nurser.

For Gas Processing Units a minimum of 2 nos. of multipurpose fire tender shall be provided.

Note:- For small petrochemical, follow (a) clause of this section.

13.2 Other Mobile Equipment

- 13.2.1** In addition to fixed monitors provided in the tank farm, following additional mobile equipment's should be provided:



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- i) Minimum 2 nos. of foam tank trailers preferably with field adjustable variable flow water cum foam monitors having foam tank capacity of 500-1000 liters and monitors capacity of minimum 1000 GPM (228 m³/hr) and which are listed or approved by standards such as BIS, UL, and FM etc.
- ii) Minimum 2 nos. of trolley/trailer mounted water cum foam monitors of capacity of minimum 2000 GPM (456 m³/hr) with field adjustable variable flow and which are listed or approved by national or international standards such as BIS, UL and FM etc. Foam induction to the monitor shall be possible from minimum 60M distance from the monitor.
- iii) 1 to 2 nos. of portable monitor of capacity ranging from 1800 to 2250 lpm at discharge pressure of 7 kg/cm².

13.2.2 For Gas processing plants, in addition to fixed monitors provided in the tank farm, following additional mobile equipment should be provided; namely:-

- (i) Minimum 2 nos. of Trolley mounted water cum foam monitors of capacity of minimum 2000 GPM (456 m³/hr) with field adjustable variable flow which are listed or approved by national or international standards such as BIS, UL and FM etc. Foam induction to the monitor shall be possible from a minimum 60 m distance from the monitor.
- (ii) 1 to 2 numbers. of portable monitor of capacity ranging from 1800 to 2250 lpm at discharge pressure of 7 kg/cm².

13.3 OTHER FIRE FIGHTING EQUIPMENT

Following, other firefighting equipment shall be provided:

- i) Emergency rescue equipment like cutters, expanders, inflatable lifting bags, leak pads, protective clothing, breathing apparatus and trolley mounted BA set
- ii) Fire Hoses: IS 636: Synthetic hose of Type 3.

The hose length shall be calculated as follows:

- a. For installation with hydrants up to 100 Nos: - One 15 m hose length/hydrant.
- b. For installation with more than 100 hydrants:
 - One 15-meter hose length/hydrant, for the first 100 hydrants; and,
 - One 15 M hose length for every 10 hydrants above 100.
- iii) Fire jeep (s) with two-way radio communication facility and towing facility.
- iv) One ambulance fitted with medical aid and suitable arrangements.
- v) Other accessories, foam making branch pipes, nozzles, etc. as per requirements.
- vi) 4" or higher diameter Hoses of suitable length for feeding to large capacity monitors wherever required.

In addition to above, provision of following equipment should also be considered:

- i) Suitable equipment for fighting high elevation fire.
- ii) Multipurpose firefighting tender should-be used as a single self-sufficient unit of having capacity of discharging foam, water/ *water mist* & DCP and thus performing multiple functions effectively, individually, or together, saving power and time in combating a fire. Such a Multipurpose firefighting tender should be used in lieu of one foam tender/ DCP tender as described in 13.1(b)-(i). Broad specifications of the firefighting equipment listed above are given in OISD-GDN-115 and Annexure-III.

14.0 STORAGE OF FIRE FIGHTING AGENTS



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The following quantities of firefighting agents shall be stored as given below in the table.

S.No	Description	Quantity to be stored
i)	Dry chemical powder: Reaction product of urea and Potassium bicarbonate (i.e. Potassium Carbamate or Potassium Allophonate base) or Siliconized Potassium bicarbonate DCP powder / Mono-ammonium phosphate based - as spare charge for the DCP tender/Trolleys. The DCP product should be approved / listed by national / International agencies such as BIS, UL, EN etc.	4000 kg for the DCP tender plus 500 kg for additional requirement. This is in addition to the charge loaded on tender.
ii)	Siliconized Potassium bicarbonate DCP powder/ Mono-ammonium phosphate based DCP powder for recharging of fire extinguishers. The DCP product should be approved / listed by national / International agencies such as BIS, UL, EN etc.	As required based on shelf life. However, minimum 10% of the total charge in the extinguishers should be maintained.
iii)	Foam compound: The foam concentrate shall be of type tested by protocols developed by National and international agencies such as BIS, UL, EN etc.	Higher of the quantities in: 53,778 liters (excluding that in foam tender) for installation having largest tank of diameter as 60m (for 3% concentrate). 93,230 liters (excluding that in foam tender) for installation having largest tank of diameter as 79m (for 3% concentrate) or (as per 6.10). - In an installation where the largest tank diameter is less than 60m, foam quantity requirement (excluding that in foam tender) shall be calculated as per Clause 6.9.) (Refer Annexure- II and Annexure-V for typical calculation for foam requirement)

15.0 DETECTION SYSTEM AND ALARM

Human beings are excellent fire detectors since they possess a sense of smell, sight, and touch. However, human senses are also unreliable due to the need for frequent rest and relaxation, a number of mechanical,



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electrical, and electronic devices have been developed to mimic human senses in the detection of gases / smoke / heat and flame in accordance with NFPA 72.

A flammable gas detector is designed to give warning about presence of flammable gas or vapor in the air, well before they reach explosive concentrations. Normally, the detector provides visible and audible alarm signals, but frequently it performs a further function by initiating control action such as increasing ventilation or shutting off the source of gas. A flammable gas detector can also be used for tracing leaks and checking that vessels or tunnels are gas free before entering.

15.1 AREAS TO BE COVERED WITH DETECTORS

15.1.1 The areas shall be covered with hydrocarbon/Hydrogen/Toxic gas detectors as per Annexure-IX.

The exact location and number of detectors should be decided on a need basis.

15.1.2 Following areas shall be provided with Flame / Heat detectors with alarm and/or system to actuate relevant fire suppression system in addition to HC detector as mentioned in 15.1.1:

- LPG spheres
- LPG filling sheds
- LPG pumps/compressors
- LPG loading/unloading, both in tank truck and tank wagon gantry
- Flammable Gas Skids / compressors
- TEAL storage and handling

15.1.3 Hydrocarbon detectors shall be installed near all potential leak source of class-A liquid hydrocarbon. e.g. tank dykes, tank manifolds, pump house manifold etc.

15.1.4 TOXIC GAS DETECTION:

- i) Toxic gas detectors shall be installed if the risk is applicable. Toxic gas detectors shall be mounted considering the prevailing wind direction and wind speed.
- ii) Additionally, Toxic gas sensors shall be installed in HVAC air intake duct of buildings that could be affected in the event of flammable or toxic release in the surrounding area.

16.0 COMMUNICATION SYSTEM

Effective communication is an essential element in the fire protection system of any plant. The following communication systems shall be provided in the Refinery/Process plants.

i) TELEPHONE

The Fire Station Control Room shall be provided with 2 nos. of internal telephones which are exclusively meant for receiving fire/emergency calls only. These phones should have facilities for incoming calls only. For general communication a separate telephone should be provided.

Fire Station shall also have a fixed landline dedicated for emergency calls only. Hot line, for contacting mutual aid parties.



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ii) PUBLIC ADDRESS SYSTEM

The public address system shall be connected to all control rooms, administration building (all floors), all departmental heads, security etc. Telephone exchange shall control and take care of this system.

iii) A.R.P. (AIR RAID PROTECTION) SYSTEM / PAGING

An air raid communication system (with civil defence) should be provided in the control room of the fire station. The details of such a system should be worked out in association with the civil defence authorities of the area.

Alternatively, group communication system (all call system)/ alpha numerical pager system should be considered for group emergency communication.

iv) FIRE SIRENS

The Fire siren/s shall be located suitably to cover the whole area with operational control in the Fire station control room. These shall be tested at least once in a week to keep them in working condition.

Fire siren code shall be as follows:

- a) Emergency Level I – The siren code for Level I shall be decided by Entity depending upon the size and complexity of installation.
- b) Emergency Level II - A wailing siren for two minutes.
- c) Emergency Level III - Same type of siren as in the case of Level II but shall be sounded for three times at the interval of one minute that is to say (wailing siren 2 minutes + gap 1 minute + wailing siren 2 minutes + gap 1 minute + wailing siren 2 minutes) and the total duration of disaster siren shall be eight minutes.
- d) ALL CLEAR (For fire): Straight run siren for two minutes.
- e) TEST: Straight run siren for two minutes at frequency at least once a week.

v) WALKIE-TALKIE / WIRELESS

All the Fire Tenders shall be provided with a walkie-talkie/ wireless system, which will help in communicating with the people in case the other system fails. Besides, key personnel coordinating emergency operations should also be provided with walkie-talkie.

vi) Intrinsically safe mobile phone may be used as alternate communication mode.

vii) FIRE & GAS ALARM SYSTEM

The fire alarm system includes manual call points, automatic gas/smoke/heat detectors, release & inhibit switches for fire suppressing clean agent and conventional or micro-processor-based data gathering panels viz. central fire alarm panel, mimic panels & associated equipment.

Manual Call Points shall be provided at suitable locations like access point, approach roads, walkways, plant area etc. These manual call points shall activate the audio-visual alarm in the Central fire alarm panel installed in fire station and in the repeater, panel installed in the respective area control room(s). The location of these points shall be conspicuously marked on the annunciation panel for proper identification.

Wherever other marketing installations like terminal/ pipeline are coexist with refinery, configuration/extension of fire alarm signal of such installations with refinery fire control room shall be done.



17.0 INSPECTION & TESTING OF FIRE PROTECTION SYSTEM:

The fire protection equipment shall be kept in good operating condition all the time and the firefighting system shall be periodically tested for proper functioning and logged for record and corrective actions. In addition to routine daily checks/maintenance, the following periodic inspection/testing shall be ensured.

17.1 FIRE WATER TANK/RESERVOIR /FOAM TANKS

- i) Above ground fire water tanks should be inspected externally & internally as per OISD-STD- 129.
- ii) The water reservoir shall be inspected periodically, not more than 5 years to ensure the integrity of the reservoir and there is no seepage and leakage from the reservoir. During such inspection, any accumulation of mud, sand, weeds, and other undergrowth, which can reduce the capacity of the reservoir and obstruct free flow of water, shall be recorded and the required remedial action shall be taken.
- iii) The foam tanks shall be inspected every 3 years externally and shall undergo internal inspection every 10 years.

17.2 FIRE WATER PUMPS

- i) Fire water pumps should be tested in line with OISD-STD-142.
- ii) Once a month, each pump should be checked for the shut off pressure and auto start operation. Observations should be logged.
- iii) Once in six months, each pump shall be checked for performance. This can be done by opening the required number of hydrants/monitors depending on the capacity of the pump and by verifying that the discharge pressure, flow, and the motor load are in conformance with the design parameters. For flow measurement suitable devices like ultrasonic instrument should be considered.

17.3 FIRE WATER RING MAIN

The ring main should be checked once a year for leaks etc. by operating one or more pumps with the hydrant points kept closed as required to get the maximum operating pressure.

The ring main, hydrants, monitors, valves should be visually inspected every month for any pilferage, defects, and damage.

All fire main valves should be checked for operation and lubricated once in six months for fresh water and once in three months for saline/ETP water.

Thickness survey & inspection of Firewater header should be done once in three years in line with OISD-STD-142.

Segment - wise flushing of main header should be done once a year.

17.4 FIRE WATER SPRAY/SPRINKLER SYSTEM

Fixed water-cooling spray/sprinkler system shall be tested in line with OISD-STD-142.

17.5 FIXED / SEMI-FIXED FOAM SYSTEM



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Foam system on storage tanks should be tested once in 12 months. This shall include the testing of foam maker/ chamber.

The foam chamber should be designed suitably to facilitate testing of foam discharge outside the cone roof tank.

Piping should be flushed with water after testing foam system.

Thickness survey & inspection of Foam system piping should be done in line with OISD-STD-142.

17.6 CLEAN AGENT BASED EXTINGUISHING SYSTEM

The systems shall be checked in line with OISD-STD-142.

17.7 MOBILE FIRE FIGHTING EQUIPMENT AND ACCESSORIES

Foam tenders should be tested at least once a week. This should include running of pump and foam generation equipment.

All other mobile equipment should be checked, serviced, and periodically tested under operating conditions, at least once a month.

Trailer mounted pumps should be test run at least once a week.

All the fire hoses shall be service pressure tested annually at maximum working pressure of fire water network/Fire fighting vehicle.

DCP tender should be visually inspected every week. This should include checking of expelled gas.

Records shall be maintained of all maintenance, testing and remedial/ corrective actions taken wherever necessary.

17.8 DCP/ CO₂/ FOAM TYPE FIRE EXTINGUISHERS:

Inspection and testing frequency and procedure shall be in line with OISD-STD-142.

17.9 COMMUNICATION SYSTEM

Fire sirens shall be tested at least once a week. Testing of Manual call points once in a six month, walkie-talkies every week and other communication systems covered under item 16.0 should be done as per manufacturer's guidelines.

17.10 DETECTORS

Calibration of gas detectors using test gas should be done once in every six months or as per manufacturer's specification, whichever is earlier.

18.0 FIRE STATION / CONTROL ROOM

The Fire Station Control Room is of critical importance as it is the main coordinating centre between the emergency site and response crew. Important activities of the control room are communication, mobilization, up-keeping, and maintenance. The location and construction of the control room should therefore be suitable for these activities.



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Fire station(s) equipped with state of art infrastructure, alarm and communication system and critical resources shall be provided at strategic locations for quick mobilization during any emergency. Entity to define the response time and ensure that all hazardous locations are within the defined response time and shall not exceed 5 minutes.

Based on the size and complexity of the entity, the number of fire station, emergency responders and resources shall be kept available to mitigate two simultaneous emergencies at the complex.

18.1 LOCATION

Fire station should be located at minimum risk area. It should be spaced at a safe distance from any process plant and other hazardous areas. For details refer to OISD-STD-118.

When planning for a new fire station, adequate land should be provided for parking and maneuvering of fire appliances. Also, access and exits of the building should not be obstructed by other vehicles.

The fire station control room should be close to the parking bay for fire appliances and should have a good view of vehicles parked.

Additional Fire station should be considered during any expansion for meeting the response time.

18.2 GENERAL:

- a) Fire Station should have 2 independent overhead storage tanks for foam compound storage, so that during emergency refilling is not delayed.
- b) The control room shall have portable emergency lights.
- c) Fire Station shall have facility to monitor fire water network pressure with low pressure alarm.
- d) Fire station should have facility to monitor the status of fire water pumps.
- e) Emergency power supply shall be ensured for the Fire Station & Firewater Pump House.

19.0 FIRE FIGHTING ORGANISATION

Petroleum Refinery and Petrochemical complex poses significant risk in terms of fire, toxic gas and explosion events. In spite of sound design, engineering practices and best in class safety management systems combined with approved operation, maintenance and inspection practices, the inherent nature of Refinery processes are vulnerable to major disaster leading to serious consequences and threat to employees, assets, environment, and community.

To mitigate these challenges, a team of fire and safety professionals shall be available at each site for emergency response and perform various functions of safety management systems of the entity. All team members shall be professionally qualified, trained, and competent to perform the assigned roles. Emergency response team members shall be identifiable with the help of distinct colour of uniform, turnout gear, Inherent Fire Retardant (IFR) coveralls or associated PPEs.

20.0 FIRE PROTECTION TRAINING

All the plant personnel shall be trained on fire prevention and firefighting aspects. Firefighting skill upgradation / refresher training shall be given periodically. The fire crew belonging to the firefighting department shall be given intensive training for the use of all equipment and in various firefighting methods



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for handling different types of fire in line with OISD-STD-154. The training of the fire personnel using digital technologies such as VR (Virtual Reality) or AR (Augmented Reality) should be explored.

Regular contractors and security personnel shall be given safety, fire Fighting and rescue operation training and record to be maintained in line with OISD-STD-154.

A fire training ground with the following training facilities should be set up:

- a) Trench fire simulation facilities
- b) A small open top tank fire simulation facility
- c) Pan fire simulation facility.
- d) Pipeline flange leak fire simulation facility
- e) Fire suits and breathing apparatus.
- f) Rescue Training modules

A mock fire drill should be conducted once in a month to rehearse the fire emergency procedure and to keep the firefighting team trained and alert and facilities in top order.

A layout of a typical fire training ground and a brief write-up on the facilities in the training ground are given in Annexure-IV.

For refineries, Petrochemicals & Gas Processing Plants not equipped with adequate fire training facilities, skill up gradation/ refresher training shall be periodically conducted at reputed institutes having live module firefighting training facilities.

21.0 MUTUAL AID

Refineries/process plants should have written mutual aid agreements with similar neighboring industries fully detailing the responsibilities of the members of the scheme, the procedures to be adopted, the minimum number of equipment and manpower and minimum quantity of consumables to be exchanged/loaned in line with OISD-STD-227.

A chart showing mutual aid arrangement shall be exhibited prominently at least in the fire station and Disaster Control Room.

22.0 FIRE EMERGENCY PROCEDURES

Each installation shall prepare a detailed "Fire emergency procedures" or emergency response plan manual derived by the ERDMP outlining the actions to be taken by each personnel during a major incident for use by the organization. This manual shall be available to all personnel in the installation.

23.0 PASSIVE FIRE PROTECTION MEASURES / OTHER SAFETY MEASURES

Although adequate fire protection is provided in an installation, Passive fire protection measures as indicated below should be adopted wherever required.

- a) Fireproofing of structural members



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- b) Spark Arresters and Flame Arresters
- c) Fire Separation Walls in concealed space/Electrical Substation/ transformer yard/ bays/ cable galleries.
- d) Fire Seals in underground sewer system / Flare Knock out Drums
- e) Impounding Basins/Dyke Walls
- f) Lightning Arresters
- g) Pressurization of Enclosure
- h) Venting Facilities of process equipment
- i) Electrical Relays and Fuses, earth leakage circuit breakers, neutral current circuit breaker
- j) Fire retardant coatings and tapes for cables
- k) Fire resistant low smoke insulation cable.
- l) Flame proof and flame-resistant electrical enclosure.
- m) Insulation of hot surfaces.

24.0 FIRE PROTECTION SYSTEM AUDIT

The fire protection system should be internally audited at least once in 3 years for compliance of OISD standards and statutory requirements.

25.0 ABBREVIATIONS

AFFF: Aqueous Film Forming Foam
 AR-AFFF: Alcohol Resistant Aqueous Film Forming Foam
 CCRU: Continuous Catalytic Reforming Unit
 CDU: Combined distillation unit
 DCP: Dry chemical powder
 DCU: Delayed Coker unit
 DHDS: Diesel hydro desulfurization
 DWST: Double walled storage tank
 FCCU: Fluidized Catalytic Cracking Unit
 FFFP: Film Forming Fluoro-Protein
 HVLRM: High-volume long-range monitor
 TEAL: Triethylaluminum
 TEB: Triethylborane

26.0 REFERENCES



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1. NFPA-11: Standard For Low Expansion Foam Systems
2. NFPA 12: Standard on Carbon Dioxide Extinguishing Systems.
2. NFPA 13: Standard For The Installation Of Sprinkler Systems
3. NFPA 14: Standard for the Installation of Standpipe and Hose Systems
4. NFPA 15: Standard For Installation Of Water Spray Systems
5. NFPA 17: Standard for Dry Chemical Extinguishing Systems
6. NFPA 18A: Standard on Water Additives for Fire Control and Vapor Mitigation
7. NFPA 20: Standard For Installation Of Centrifugal Fire Pumps
8. NFPA 24: Standard for the Installation of Private Fire Service Mains and Their Appurtenances
9. NFPA 30: Flammable and Combustible Liquids
10. NFPA 72: National Fire Alarm and Signaling Code
11. NFPA-2001: Standard For Clean Fire Extinguishing System
12. NFPA-855: Standard for the Installation of Stationary Energy Storage Systems
13. API-2001: Fire Protection In Refineries
14. API RP 2030: Application of Fixed Water Spray Systems for Fire Protection in the Petroleum and Petrochemical Industries
15. Petroleum Rules -2002
16. International Safe Practices Of Oil Industry
17. OISD-GDN-115: Guidelines on specification of Fire Fighting Equipment & Mitigation Strategy For Credible Fire Scenario
18. OISD-STD-118: Layouts for Oil and Gas Installations
19. OISD-STD-142: Inspection Of Fire Fighting Equipment & Systems
20. OISD-STD-154: Safety Aspects In Functional Training
21. Oil Mines Regulations, 2017



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22. Ozone Depletion Substances Regulation & Control Rules 2000, Ministry of Environment & Forests, Government of India
23. IS-15683: Portable Fire Extinguishers - Performance and Construction - Specification
24. IS:14609 : Dry Chemical Powder for fighting A,B,C Class Fire – Specification
25. IS 5: Colours for ready mixed paints and enamels
26. IS 15683: Portable Fire Extinguishers Performance and Construction Specification
27. IS 2190: Selection, installation and maintenance of first-aid fire extinguishers – Code of practice
28. BS EN 13565-2:2018: Fixed firefighting systems - Foam systems
29. UL154: Carbon-Dioxide Fire Extinguishers
30. UL299: Dry Chemical Fire Extinguishers
31. UL711: Rating & Fire Testing of Fire Extinguishers
32. UL162: Foam Equipment and Liquid Concentrates
33. T4S : Technical Standards and Specifications including Safety Standards for Petroleum Refineries and Gas Processing Plants

27.0 Conversion Table:

1 L/min	0.06 m ³ /h
1 gpm	0.228 m ³ /h

**28. ANNEXURE****ANNEXURE-I****EXAMPLE FOR CALCULATION OF FIRE WATER FLOW RATE^{Note1}****1. DESIGN BASIS**

The fire water system in installation shall be designed to meet the fire water flow requirement to fight two major fire simultaneously.

2. FIRE WATER DEMAND

Various areas which can be under fire shall be considered and fire water demand for each area shall be calculated based on design basis, as indicated below:

2.1 FLOATING ROOF TANKS PROTECTION**a) Data:**

Total storage capacity in one dyke area	=	1,20,000 m ³
No. of tanks	=	2
Capacity of each tank	=	60,000 m ³
Diameter of each tank	=	79 m
Height of each tank	=	14.4 m

b) Cooling water requirement:

Cooling water rate @ 3 lpm/ m² of tank shell area for tank on fire,

Cooling water required	=	($\pi \times 79 \times 14.4 \times 3$) lpm
	=	10,726 lpm
	=	644 m ³ /hr

Assuming that second tank is located within the same tank dyke. Therefore, in such case cooling water required at the rate of 3 lpm/ m² of tank shell area shall be 644 m³/hr.

Total cooling water = 1288 m³/hr

c) Foam water requirement for rim seal area:

Water flow is required for applying foam on a largest tank burning surface area.

For floating roof tank of 79 m diameter,

Diameter of the tank (D1)	=	79 m
Distance of foam dam from shell	=	0.8 m
Diameter of roof up to foam dam (D2)	=	79 - (2 x 0.8) = 77.4 m
the rim seal area	=	$((\pi / 4) \times (79^2 - 77.4^2))$ m ²
	=	197 m ²
Foam solution rate @ 12 lpm/ m ²	=	2,364 lpm
(For 3% foam concentrate)	=	(0.97 x 2364) lpm
	=	2293 lpm.
	=	$\frac{2293 \times 60}{1000}$ m ³ /hr
	=	138 m ³ /hr.



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Note-1: These are sample calculations only. Calculations on the basis of actual site conditions and dimensions need to be carried out for each installation as per guidelines provided in clause 5.2.

- d) Fire water for supplementary hose stream based on 4 hydrant streams + 1 High-Volume Long-Range water monitor.

$$4 \times 36 \text{ m}^3/\text{hr} + 1 \times 228 \text{ m}^3/\text{hr} = 372 \text{ m}^3/\text{hr}$$

- e) **Total water required:**

Tank cooling	1288 m ³ /hr
Foam application	138 m ³ /hr
Supplementary stream	372 m ³ /hr
Total	1798 m³/hr

2.2 CONE ROOF TANKS PROTECTION

- a) **Data:**

Total storage capacity	=	50,000 m ³
No. of tanks	=	4 with 12,500 m ³ capacity each.
Diameter of each tank	=	37.5 m
Height of each tank	=	12 m

- b) **Cooling water requirement:**

Cooling water rate	=	3 lpm/ m ² of tank shell area for tank-on-fire
Cooling water required	=	$\pi \times 37.5 \times 12 \times 3$
	=	4243 lpm
	=	255 m ³ /hr

Cooling water required for other tanks at the rate of 3 lpm/ m² of shell area for tanks in same dyke.

	=	3 x 255 m ³ /hr
	=	765 m ³ /hr
Total cooling water rate	=	(255 + 765) m ³ /hr
	=	1020 m ³ /hr

- c) **Foam water requirement** (for 1 tank only) @ 5 lpm/ m²,

Foam solution rate	=	$\frac{(\pi \times (37.5)^2 \times 5)}{4}$ lpm
	=	5525 lpm
For 3% foam concentration	=	(5525 x 0.97) lpm
	=	5359 lpm
	=	322 m ³ /hr

- d) Fire water for **supplementary hose** stream = 372 m³/hr

- e) **Total water required :**



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Tank cooling	1020 m ³ /hr
Foam application	322 m ³ /hr
Supplementary stream (including 1 HVLRM)	372 m ³ /hr
Total	1,714 m³/hr

2.3 LPG SPHERES AREA PROTECTION**a) Data:**

No. of sphere in one area	=	3
Diameter of each sphere	=	17 m

b) Cooling water requirement:

water rate for cooling	=	($\pi \times 17^2 \times 10.2$) lpm
	=	9,265 lpm
	=	556 m ³ /hr

Considering other 2 spheres located within (R+30) M from centre of sphere and fire cooling water rate for 3 spheres

$$= 1668 \text{ m}^3/\text{hr}$$

c) Hose stream requirement (including 1 HVLR) = 372 m³/hr

d) Total water requirement = **2040 m³/hr**

2.4 LPG RAIL WAGON LOADING GANTRY PROTECTION**a) Data:**

Total No. of loading points	=	Conventional or BTPN.
Width of Tank wagon gantry	=	12 m

b) Cooling water requirement:

Divide total area of gantry into equal segments such that each segment measuring 15 m X 12 m and consider 3 segments operating at a time.

Water rate required	=	(3 x 15 x 12 x 10.2) lpm
	=	5508 lpm
	=	330 m ³ /hr

Water Requirement for supplementary Hose:

Water for 4 single hydrant streams	=	4 x 36 = 144 m ³ /hr.
Water for 1 monitor stream (HVLR)	=	1 x 228 = 228 m ³ /hr.
Total water requirement	=	372 m ³ / hr.

Total water flow rate for gantry protection

(a) Gantry cooling	=	330 m ³ /hr.
(b) Supplementary hose requirement	=	372 m ³ / hr
Total	=	702 m³/hr.



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2.5 PROCESS UNIT PROTECTION

For process unit protection in case of fire, water is to be applied using fixed water monitors and hose lines. Unit blocks separation is by 30 meter. Three following alternatives are considered for fire water rate.

Alternate-I

Total unit area = 120 x 80 m²

Consider water rate @ 1 lpm/ m² on area basis,

Water rate = (9600 x 1) lpm
= 576 m³/hr

Water for supplementary hose stream (including 1 HVLR) = 372 m³/hr

Total water rate = **948 m³/hr**

Alternative - II

Consider a 10m x 10m portion of process unit area on fire. Provide water cover over an area of 30m x 30m at the rate of 10.2 lpm/ m²,

Water rate = (900x10.2) lpm
= 9180 lpm
= 551 m³/hr

Water for supplementary hose steam (including 1 HVLR) = 372 m³/hr

Total water rate = **923 m³/hr**

Alternate – III

Water required for portion of unit area provided with fixed spray system (Extreme Hazardous Area)

Area assumed = 1000 m²
Water rate = 10.2 lpm/ m²
Cooling water required = 10200 lpm
= 612 m³/hr

Water for supplementary hose stream (including 1 HVLR) = 372 m³/hr

Total cooling water required = **984 m³/hr**

Considering the maximum water under alternative I, II & III

Design flow rate = **984 m³/hr**

2.6 Fire water calculation for full surface fire on largest floating roof tank (roof sinking case) – Treated as a single contingency**a) Data:**

Total storage capacity in one dyke area = 120,000 m³

No. of tanks = 2

Capacity of each tank = 60,000 m³

Diameter of each tank = 79 m

Height of each tank = 14.4 m



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b) Cooling water requirement:Cooling water rate @ 3 lpm/ m² of tank shell area for tank-on-fire

$$\begin{aligned}
 \text{Cooling water required} &= \pi \times 79 \times 14.4 \times 3 \\
 &= 10726 \text{ lpm} \\
 &= 644 \text{ m}^3/\text{hr}
 \end{aligned}$$

Assuming that second tank is located within the tank dyke at a distance more than 30m from the tank shell.

Then, cooling water requirement @ 3 lpm/ m² of tank shell area shall be 644 m³/hr.

$$\begin{aligned}
 \text{Total cooling water} &= (644 + 644) \text{ m}^3/\text{hr} \\
 &= \mathbf{1288 \text{ m}^3/\text{hr}}
 \end{aligned}$$

c) Water requirement in foam application

$$\text{Foam Application Rate} : 9.75 \text{ lpm}$$

$$\begin{aligned}
 \text{Foam Solution Requirement} &= ((\pi \times 79 \times 79) / 4 \times 9.75) \text{ lpm} \\
 &= 47810 \text{ lpm} \\
 &= 2868 \text{ m}^3/\text{hr}
 \end{aligned}$$

$$\begin{aligned}
 \text{Water required for the foam solution(3\%)} &= 97\% \times 2868 \text{ m}^3/\text{hr} \\
 &= \mathbf{2782 \text{ m}^3/\text{hr}}
 \end{aligned}$$

d) Fire water for supplementary hose stream based on 4 hydrant streams + 1 High-Volume Long-Range water monitor.

$$4 \times 36 \text{ m}^3/\text{hr} + 1 \times 228 \text{ m}^3/\text{hr} = 372 \text{ m}^3/\text{hr}$$

e) Total water required:

$$\begin{aligned}
 \text{Tank cooling} &1288 \text{ m}^3/\text{hr} \\
 \text{Foam application} &2782 \text{ m}^3/\text{hr} \\
 \text{Supplementary stream} &372 \text{ m}^3/\text{hr} \\
 \text{Total} &\mathbf{4442 \text{ m}^3/\text{hr}}
 \end{aligned}$$

$$\text{Say Total water requirement} = \mathbf{4500 \text{ m}^3/\text{hr}}$$

3.0 TOTAL DESIGN FIRE WATER RATE

For two major fire fought simultaneously

Fire water rates for 5 cases are given below:

- i) Floating roof tank protection = 1798 m³/hr
- ii) Cone roof tank protection = 1714 m³/hr
- iii) LPG sphere protection = 2040 m³/hr
- iv) LPG rail wagon loading gantry = 702 m³/hr
- Protection
- v) Process unit protection = 984 m³/hr



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- 3.1** For fighting the above two major fire simultaneously, the design firewater rate is the sum of the two highest water rates i.e.

$$\begin{aligned}
 \text{Design fire water rate} &= 2040 \text{ m}^3/\text{hr} + 1798 \text{ m}^3/\text{hr} \\
 &= \mathbf{3838 \text{ m}^3/\text{hr}} \\
 \text{Say} &= \mathbf{3850 \text{ m}^3/\text{hr}}
 \end{aligned}$$

- 3.2** For full surface fire of largest floating roof tank (Roof sinking case):
Total firewater flow rate required as per typical calculations shown at 2.6 is 4500 m³/hr.

The design Firewater rate shall be **highest of above 3.1 or 3.2.**

NOTE:

Full surface fire of a floating roof tank roof sinking case being a remote possibility, it is considered as a single largest contingency for the purpose of arriving at design fire water requirement.

**ANNEXURE - II****EXAMPLE FOR CALCULATION OF FOAM COMPOUND REQUIREMENT****1. CONE ROOF TANK PROTECTION:****i) Data :**

Total Storage capacity in one dyke area	=	50000 m ³
Number of tanks	=	4
Diameter of each tank	=	37.5 m
Height of each tank	=	12 m

ii) The quantity of foam compound shall be calculated as follows:

Consider foam solution application @ 5 lpm/ m² for the liquid surface of the single largest cone roof tank in the dyke area.

$$\begin{aligned}
 \text{Foam solution rate} &= \frac{\pi \times (37.5)^2}{4} \times 5 \\
 &= 5525 \text{ lpm} \\
 \text{Foam compound required (3\%)} &= (5525 \times 3 / 100) \text{ lpm} \\
 &= 166 \text{ lpm} \\
 \text{Foam compound quantity for 65 minutes} &= 166 \times 65 \\
 &= \mathbf{10,790 \text{ liter}}
 \end{aligned}$$

iii) Consider one foam monitor of 3785 lpm foam solution capacity:

$$\begin{aligned}
 \text{3\% Foam compound required} &= \mathbf{114 \text{ lpm}} \\
 \text{Foam compound required for 65 minutes} &= \mathbf{7410 \text{ liter}}
 \end{aligned}$$

iv) Consider 2 hose streams of foam with a capacity of 900 lpm of foam solution capacity

$$\begin{aligned}
 \text{3\% Foam compound required} &= \mathbf{54 \text{ lpm}} \\
 \text{Foam compound required for 65 minutes} &= \mathbf{3510 \text{ liter}}
 \end{aligned}$$

v) Total foam compound required for cone roof tank area Protection:

$$\begin{aligned}
 \text{Foam compound required for Cone Roof Tank} &= 10,790 \text{ liter} \\
 \text{Foam Compound required for 1 Foam Monitor} &= 7,410 \text{ liter} \\
 \text{Foam Compound required for 2 hose streams} &= 3510 \text{ liter} \\
 \text{Total} &= 21,710 \text{ liter}
 \end{aligned}$$

$$\text{Say} = \mathbf{22,000 \text{ liters}}$$



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2. FLOATING ROOF TANK PROTECTION

i) Data:

Total Storage Capacity in one dyke	=	1,20,000 m ³
No. of Tanks	=	2
Capacity of Each Tank	=	60,000 m ³
Diameter of each tank	=	79 m
Height of each tank	=	14.4 m

iii) Consider foam solution application rate of 12 lpm/ m² of seal area of the single largest floating roof tank in the dyke area :

For floating roof tank of 79 m diameter,

Diameter of the tank (D1)	=	79 m
Distance of foam dam from shell	=	0.8 m
Diameter of roof up to foam dam (D2) =		(79- (2X0.8)) m = 77.4 m
Rim seal area	=	((π /4) x (79 ² -77.4 ²)) m ²
	=	197 m ²

Foam solution rate @ 12 lpm/ m² = 2364 lpm

3% Foam Compound required = 70.9 lpm

Foam Compound required for 65 mins. = 4609 liter

iii) Foam Compound required for 1 foam monitor and 2 hose streams as calculated for cone roof protection

1 Foam monitor	7410 liter
2 Hose streams	3510 liter

iv) Total foam compound required for floating roof tank area Protection:

Foam Compound required for Floating Roof Tank	4609 liter
Foam compound required for 1 foam monitor	7410 liter
Foam compound required for 2 hose streams	3510 liter
Total required	15529 liter

Say, 16,000 liter

3. On the lines of the above example foam compound requirement should be calculated for various dyke areas. Requirements to fight major fire in two dyke areas (with maximum foam compound rates requirements) should be added, to arrive at the total requirement of the installation.

For example, for 2 cone roof tank dyke areas with largest tank diameter of 37.5 meters in each area, foam compound required works out as 2x22000 liters i.e. 44,000 liters.

Similarly for 2 floating roof tank dyke areas with largest tank diameter of 79 M. in each area, foam compound required works out as 2 X 16000 liters i.e. 32,000 liters.

4. Foam Requirement for Full surface fire of the largest floating roof tank (roof sinking Case): considered as a single largest contingency & detailed at Annexure-V



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ANNEXURE - III**BROAD SPECIFICATIONS FOR FIRE FIGHTING EQUIPMENT
(Refer OISD STD -115 for details)**

No.	Equipment	Broad Specification Capacity And Size	Suitability
I.	Foam Tender	- Diesel Engine Chassis with compatible PTO Unit - 19 ton payload - Foam compound Tank - 3000 liters. - Water tank - 4000 liters. - Pump capacity - 4000 lpm@ 8.5 kg/sq.cm g & 2500 liters/min @ 10 kg/sq.cm g - Foam/water monitor of capacity. 2500 to 2800 liter/min – 1 No.	- To supply Foam Solution to Fixed foam System. - To supply foam to spill fires.
II.	Hydraulic Platform with articulated or telescopic arm	- Diesel Engine Chassis with compatible PTO Unit - MIN.HP280 - Foam generating equipment. - Cage area – 1.4 sq. mt. - Extinguishing Articulated boom Of 30 M height - Throwing distance Horizontal :80-100m Height: 65-75mt. - Boom movement – 360 degrees. - Foam/water monitor capacity. 5000 to 6000 lpm - Pump capacity – 5000 to 6000. Lts/min. at 10 kg/sq. cm.g - Remote operation from rear of truck - Operation to suit water. Connection from hydrants	- For foam application on oil fires - For fire fighting Of tall columns And inaccessible area - For rescue
III.	DCP Tender	- Diesel engine chassis –payload :13 - 2 vessels of 2000 kg DCP each - Expellant gas system (Nitrogen gas cylinders 28x50 lts at 140 kg/cm² - Automatic regulator for output and range - DCP monitor with range 15, 25 and 40 kg/second at operating pressure of	- For gas Fires and Spill fires



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		14 kg/cm ²	
IV.	Emergency Rescue Tender	- Monitor throw - 40 to 50 mtrs - Rescue equipment. - Breathing equipment - Protective clothing - First Aid Equipment	- For rescue Operations
V.	Foam Tank Trailers with Water-cum- Foam monitors.	- Foam compound storage capacity 1000 litres - Filed adjustable variable flow Water cum foam monitor of capacity 1000 GPM which are listed or approved by national or international standards such as BIS, UL and FM etc.	- For spill fires
VI.	Trolley mounted. Water-foam- Monitors	- Field adjustable variable flow Water cum foam monitor of capacity 2000 GPM which are listed / approved by national or international standards such as BIS, UL and FM etc. Foam induction to the monitor shall be possible from minimum 60M distance from the monitor.	- For oil fire - For hot work
VII.	Foam Nurser (Trailer mounted foam supply tank)	- Foam Compound storage - 8000 litres. - Pump capacity: 160 liters/min	- To supply Foam compound During fire fighting
VIII.	Mobile fire Trailer pump	- Pump capacity 1800 to 2250 Lts/min at 7 kg/sq.cm. g	- As a booster Pump - Fire water use from additional source
IX.	Fire hoses -2 ½"	- Reinforced rubber lined hoses type 3 as per IS 636	- General fire Fighting
X.	Fire jeep	- Standard design with two-way radio Communication facility	- Towing trailer Type equipment - General vigilance
XI.	Ambulance	- Medical first aid equipment	- Medical aid



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- | | | | | | |
|------|---|---|--------------------------------|---|-------------------|
| XII. | Other accessories | - | As per standard specifications | - | For fire Fighting |
| | Foam making
branch pipes,
nozzle etc. | | | | |

NOTE: The capacities of equipment given above are the minimum required. However, higher capacities can be provided on need basis.

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**ANNEXURE - IV****FIRE TRAINING GROUND - FACILITIES****1.0 LAYOUT OF TYPICAL FIRE TRAINING GROUND**

A typical layout of a fire training ground is shown in fig.1

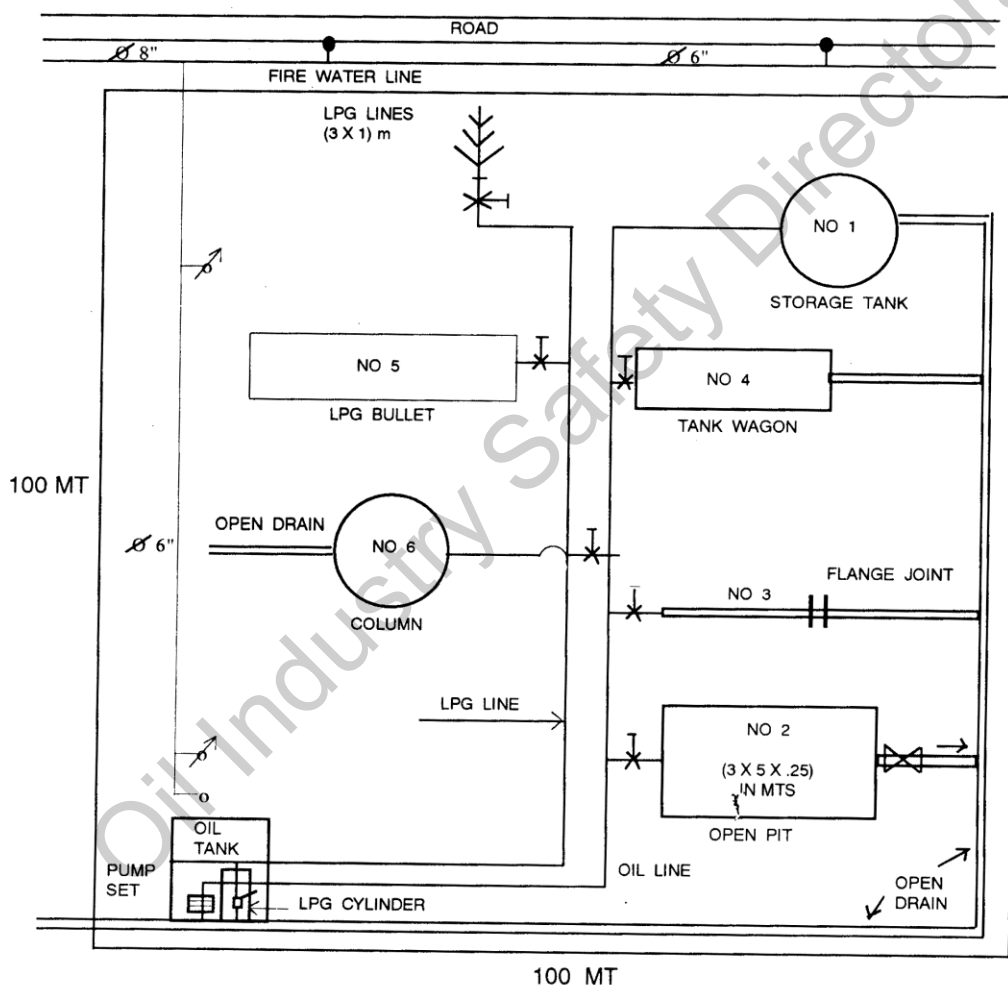


Fig. 1
LAYOUT OF TYPICAL FIRE TRAINING GROUND

2.0 DETAILS OF FACILITIES IN TRAINING GROUND



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Details of different facilities in the fire training ground are described below.

2.1. FACILITY FOR STORAGE TANK FIRE

Storage tanks are provided with fixed foam line connection and fixed water spray system to fight tank fires. The foam is to extinguish the fire, and water spray is to prevent spreading of fire in the neighboring tanks. Training ground should have a tank with fixed foam installation with isolation valves on product lines.

2.2 FACILITY FOR PIT FIRE:

During operation of the refinery, sometimes hydrocarbon liquids get released from the process equipment and accumulate in the open areas/ pit. To fight such fires, the facility should be provided on the ground.

2.3 FACILITY FOR PIPELINE FIRES:

In the refinery equipment are connected by pipelines and to run down the products from the process units to storage tanks. These pipes can leak at flange joints and result in a fire. Facilities should be provided to simulate such fires.

2.4. FACILITIES FOR WAGON GANTRY FIRE:

Gantry fires are difficult to control due to rapid spreading of fires all around the gantry. Such fires can be controlled with the following arrangements

- a) Isolate the oil supply
- b) Cooling of the area as well as surroundings with water spray & facilities should be provided for such simulations.

2.5 FACILITIES FOR LPG FIRE

LPG is produced in the refinery and stored in containers from where it is dispatched to the market. Chances of fires due to leakage of LPG in the process unit are less as it is processed in closed system. Probability of LPG fires during bottling; loading/unloading of cylinders is more.

The most dangerous aspect in LPG fires is explosion of cylinder. To tackle such fires, training facility should be installed.

2.6 FACILITY FOR HIGH ELEVATION COLUMN FIRE

In the refinery, there are tall structures such as columns/vessels. To tackle fires on such equipment the most important activities are isolation and cooling the equipment as well as surrounding area. Generally, isolation of inlet side of the equipment is not very difficult, most difficult isolation is on the outlet side. To approach to the isolation valve, personal protective equipment with water screen is necessary. Facilities should be provided to simulate all such actions.

**ANNEXURE-V****Example of Fire Case in a large Floating Roof Tank after sinking of floating roof**

Example for calculation of Foam Requirement for Floating Roof tank with Portable Monitors:

DATA:

1. Diameter of Tank = 79 m
2. Type of Roof = Floating Roof
3. Foam Application Rate = 9.75 lpm

$$\text{Foam Solution Requirement} = \frac{\pi \times 79 \times 79}{4} \times 9.75$$

$$= 47,810 \text{ lpm}$$

$$= 2868 \text{ m}^3/\text{hr}$$

$$\text{Say,} = 2,900 \text{ m}^3/\text{hr}$$

This much quantities has to be thrown over to sunken roof area with the help of external long-range high-volume monitors from the roadside periphery of the tank farm. The same can be achieved by 5 nos. of 2000 gpm such monitors. In design firewater rate calculation in Ann-I, sinking of floating roof *of the largest tank* has been considered as a single largest contingency to sizing the fire water network around tank farms to take up such firewater load so that long range monitors can be fed from this network by diverting water available in other areas of the installation to tank farms in such emergency.

$$\begin{aligned} \text{Foam Compound Requirement(3\%)} &= 47810 \times 3 / 100 \\ &= 1434 \text{ lpm} \end{aligned}$$

$$\begin{aligned} \text{Foam Compound Requirement} &= (1434 \times 65) \text{ liter} \\ \text{for 65 minutes with 3\% concentration} &= 93,230 \text{ liters} \end{aligned}$$



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ANNEXURE-VI**Fire water and Foam calculation for full surface fire on largest floating roof tank (roof sinking case) – Treated as a single contingency****a) Data:**

Total storage capacity in one dyke area	=	120,000 m ³
No. of tanks	=	2
Capacity of each tank	=	60,000 m ³
Diameter of each tank	=	79 m
Height of each tank	=	14.4 m

b) Cooling water requirement:Cooling water rate @ 3 lpm/ m² of tank shell area for tank-on-fire

$$\begin{aligned}\text{Cooling water required} &= \pi \times 79 \times 14.4 \times 3 \\ &= 10726 \text{ lpm} \\ &= 644 \text{ m}^3/\text{hr}\end{aligned}$$

Assuming that second tank is located within the tank dyke at a distance more than 30m from the tank shell.

Then, cooling water requirement @ 3 lpm/ m² of tank shell area shall be 644 m³/hr.

$$\begin{aligned}\text{Total cooling water} &= (644 + 644) \text{ m}^3/\text{hr} \\ &= 1288 \text{ m}^3/\text{hr}\end{aligned}$$

c) Water requirement in foam application

$$\begin{aligned}\text{Foam Application Rate} &: 9.75 \text{ lpm} \\ \text{Foam Solution Requirement} &= ((\pi \times 79 \times 79) / 4 \times 9.75) \text{ lpm} \\ &= 47810 \text{ lpm} \\ &= 2868 \text{ m}^3/\text{hr} \\ \text{Water required for the foam solution(3\%)} &= 97\% \times 2868 \text{ m}^3/\text{hr} \\ &= 2782 \text{ m}^3/\text{hr}\end{aligned}$$

d) Fire water for supplementary hose stream based on 4 hydrant streams + 1 High-Volume Long-Range water monitor.

$$4 \times 36 \text{ m}^3/\text{hr} + 1 \times 228 \text{ m}^3/\text{hr} = 372 \text{ m}^3/\text{hr}$$

e) Total water required:

Tank cooling	1288 m ³ /hr
Foam application	2782 m ³ /hr
Supplementary stream	372 m ³ /hr
Total	4442 m³/hr

$$\text{Say Total water requirement} = 4500 \text{ m}^3/\text{hr}$$

Foam Concentrate (3%) Required for 65 Minutes continuous discharge = 93,230 Liters (93.2 M³) (reference Annexure-V)

**ANNEXURE-VII****SYSTEM OF AUTOMATIC ACTUATED RIM SEAL FIRE DETECTION AND EXTINGUISHING SYSTEM FOR EXTERNAL FLOATING ROOF TANKS STORING CLASS-A PETROLEUM**

The automatic actuated flooding system is designed to automatically detect and extinguish the floating roof tank rim seal fire at its incipient stage. The minimum requirement for the design of the system is given below:

1.0 EXTINGUISHING MEDIA :

Film Forming Fluoroprotein Foam (FFFP) / Aqueous Film Forming Foam (AFFF) / Alcohol resistant Aqueous Film Forming Foam (AR-AFFF) type concentrate is used in the system.

1.1 FOAM APPLICATION SYSTEM :

A large storage tank requires one or more than one modular units for foam application in the entire rim seal. Each such unit consists of a foam distribution pipe, laid along the tank perimeter over the rim seal area. The spray nozzles for foam application are mounted on the distribution pipe at suitable intervals. Distribution pipe is permanently connected to a storage vessel containing pre-mix foam. The premix foam solution is kept pressurized with nitrogen / CO₂ in a vessel. The system is designed for a minimum foam application rate of @ 18 lpm/ m² of rim seal area. For effective control, foam is discharged in approximately 40 seconds.

1.2 ALARM & AUTO-ACTUATION SYSTEM:

In case of fire on the rim seal, it is automatically detected by a device capable to sense the same. The device then actuates the spray system for application of foam in the complete area of rim seal to quickly extinguish the fire in its incipient stage. An audio-visual alarm is also coupled with the detection & extinguishing system for necessary fire alert.

The system includes a fire detection system which senses fire and actuates the automatic release of the extinguishing medium on the rim seal area. Each tank shall have independent detection & extinguishing system.

The validity of the approach must be demonstrated by the designer for an effective total flooding extinguishing system which quickly detects and extinguishes fire in its incipient stage without re-flash. Also, the design considerations should include the impact of the weight of the modules placed on the floating roof.

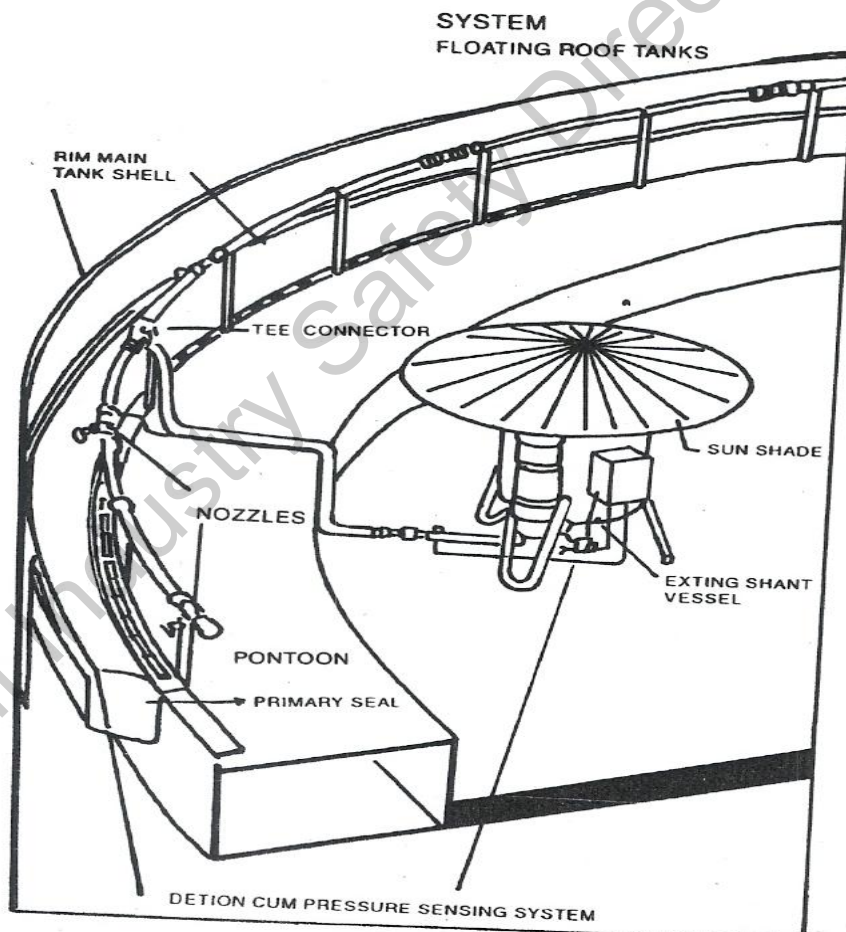
The detection system needs to be highly reliable and shall work at varied site ambient temperatures for protection of rim seal fire.

These detection systems shall be listed and/ or approved by any of the national or international standards such as agencies like BIS, UL and FM etc.



1.3 CALCULATIONS FOR MODULAR FOAM APPLICATION SYSTEM FOR 79- METER DIA TANK:

- Rim seal area of Tank : $\pi \times 79 \times 0.3 = 74.5 \text{ m}^2$ (Considering a flexible seal area of typically 300 mm)
 - Rate of Foam application @ 18 LPM/ m² = 1341 LPM
 - Total Foam solution required in 40 sec = 894 liter
 - Total nos. of Modular unit required for the tank = 7 *
- *(considering a vessel of 150-liter capacity containing 135 liter of Foam)



**ANNEXURE-VIII****BRIEF DESCRIPTION OF FIRE FIGHTING FOAM****1.0 FIRE FIGHTING FOAM:**

Firefighting foam is a homogeneous mass of tiny air or gas filled bubble of low specific gravity, which when applied in correct manner and in sufficient quantity, forms a compact fluid and stable blanket which is capable of floating on the surface of flammable liquids and preventing atmospheric air from reaching the liquid.

2.0 TYPES OF FOAM COMPOUND

Two Types of foams are used for fighting liquid fires:

2.1 CHEMICAL FOAM:

When two or more chemicals are added the foam generates due to chemical reaction. The most common ingredients used for chemical foam are sodium bicarbonate and aluminum sulphate with stabilizer. The chemical foam is generally used in Fire extinguishers.

2.2 MECHANICAL FOAM:

It is produced by mechanically mixing gas or air to a solution of foam compound (concentrate) in water. Various types of foam concentrates are used for generating foam, depending on the requirement and suitability. Each concentrate has its own advantage and limitations. A brief description of foam concentrates is given below.

3.0 MECHANICAL FOAM COMPOUND:

Mechanical foam compound is classified into 3 categories based on its expansion ratio.

3.1 LOW EXPANSION FOAM: Foam expansion ratio can be up 20 times (20 : 1)

The low expansion foam contains more water and is better resistant to fire. It is suitable for hydrocarbon liquid fires and is widely used in oil refinery, oil platforms, petrochemical and other chemical industries.

3.2 MEDIUM EXPANSION FOAM: Foam expansion ratio vary from (21: 1 to 200: 1) as produced by self-aspirating foam branch pipes with nets. This foam has limited use in controlling hydrocarbon liquid fire because of its limitations w.r.t. poor cooling, poor resistant to hot surface/radiant heat etc.**3.3 HIGH EXPANSION FOAM:** Foam expansion ratio vary from (201: 1 to 1000: 1). This foam has also very limited use in controlling hydrocarbon liquid fire because of its limitations w.r.t. poor cooling, poor resistant to hot surface/radiant heat etc. It is used for protection of hydrocarbon gases stored under cryogenic conditions and for warehouse protection.



4.0 TYPES OF LOW EXPANSION FOAM:

Foam concentrates to be used shall conform to IS 4989-2018 and amendments thereof /UL-162.

4.1 PROTEIN BASE FOAM:

The foam concentrate is prepared from hydrolyzed protein either from animals or vegetable source. The suitable stabilizer and preservatives are also added.

The concentrate forms a thick foam blanket and is suitable for hydrocarbon liquid fires, but not on water miscible liquids. The effectiveness of foam is not very good on deep pools or low flash point fuels which have had lengthy pre-burn time unless applied very gently to the surface.

The concentrate is available for induction rate of 3 to 6%. The shelf life of concentrate is 2 years.

4.2 FLUORO PROTEIN FOAM:

This is similar to protein base foam with fluoro-chemical which makes it more effective than protein base foam.

The concentrate forms a thick foam blanket and is suitable for hydrocarbon liquid fires, but not on water miscible liquids. The foam is very effective on deep pools of low flash point fuels which have had lengthy pre burn time.

The concentrate is available for induction rate of 3 to 6% and the shelf life is similar to that of protein base foam.

4.3 AQUEOUS FILM FORMING FOAM (AFFF):

The foam concentrate mainly consists of fluoro carbon surfactants, foaming agent and stabilizer. This can be used with fresh water as well as with sea water.

It produces very fluid foam, which flows freely on liquid surface. The aqueous film produced suppresses the liquid vapour quickly. The foam has quick fire knock down property and is suitable for liquid hydrocarbon fires. As the foam has poor drainage rate, the effectiveness is limited on deep pool fires of low flash point fuels which have lengthy pre burn time.

The concentrate is available for induction rate of 1% to 6% and the shelf life is more than 15 years. This can also be used with non-aspirating type nozzles.

4.4 MULTIPURPOSE AR-AFFF:

Multipurpose AR-AFFF concentrate is synthetic, foaming liquid designed especially for fire protection of water-soluble solvents and water insoluble hydrocarbon liquids. This can be used either with fresh water or sea water.

When applied it forms foam with a cohesive polymeric layer on liquid surface, which suppresses the vapour and extinguishes the fire. The foam is also suitable for deep pool fire because of superior drainage rate and more resistive to hot fuels/radiant heat.

The shelf life of concentrate is not less than 10 years. This can also be used with non-aspirating type nozzles.



4.5 FILM FORMING FLOURO PROTEIN (FFFP)

FFFP combines the rapid fire knock down quality of conventional film forming AFFF with the high level of post fire security and burn back resistance of flouro protein foam. The concentrate can either be used with fresh water or sea water.

The foam is suitable for hydrocarbon liquid fires including deep pool fires of low flash point fuels which have had lengthy pre burn time.

The concentrate is available for induction rate of 3 to 6% and the shelf life is not less than 5 years. This can also be used with non-aspirating type nozzles.

5.0 TYPES OF MEDIUM AND HIGH EXPANSION FOAM:

Synthetic foam concentrate is used with suitable devices to produce medium and high expansion foams. This can be used on hydrocarbon fuels with low boiling point. The foam is very light in weight and gives poor cooling effect in comparison to low expansion foams. The foam is susceptible to easy breakdown by hot fuel layers and radiant heat.

The induction rate in water can vary from 1.5 to 6%.

Many of the low expansion foam concentrate can also be used with suitable devices to produce medium / high expansion foam.



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ANNEXURE-IX**DETECTOR LIST**

S.NO.	LOCATION NEAR	TYPE OF GAS DETECTOR
1.	Battery limit of process unit	HC
2.	Lighter HC pumps	HC
3.	HC drainage check point	HC
4.	Fuel Gas knock out drum (KOD)	HC
5.	Flare knock out drum (KOD)	HC
6.	Acid Gas Flare knock out drum	H ₂ S
6.	Suction side of forced draft air blower	HC
7.	C4 and lighter gas pumps	HC
8.	Breech lock exchanger area	HC & H ₂ S
9.	Air fin cooler area	H ₂ S & HC
10.	Hydrogen Gas compressor	H ₂
11.	Recycle gas compressor(Hydro processing units)	H ₂ &H ₂ S
12.	Incinerator	SO ₂ & H ₂ S
13.	SRU unit	SO ₂ & H ₂ S
14.	BOD sump/ ETP oil catcher	HC & H ₂ S
15.	Sour water pump	H ₂ S



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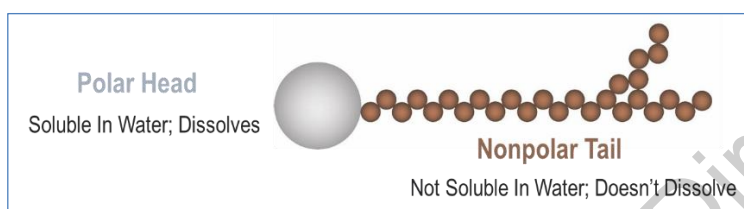
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16.	Rich amine pump	H ₂ S & HC
17.	LPG bulk truck loading area/LPG bulk wagon loading area/LPG bottling, storage, repair sheds	HC
18.	LPG Horton sphere	HC
19.	Air-intake point for control room	HC
20.	Air compressor air suction point	HC
21.	Process cooling tower sump and cell area	HC
22.	Pipeline Culverts	HC
23.	Computer room, Server room, Process control rooms, Record room	Suitable type
24.	Unmanned electric substations / MCC rooms	Suitable type
25.	Cable galleries	Suitable type
26.	Chemical Storage	Suitable type

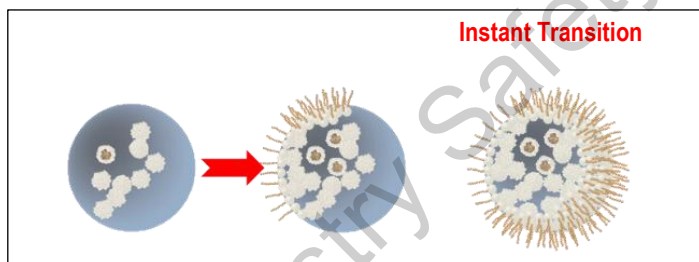
**ANNEXURE-X****Encapsulator Agent**

Encapsulator Agents are a type of water additive whose basic building blocks are spherical micelles. While most of the conventional firefighting media separate fuel from oxygen, Encapsulator Agents work on a molecular level. Encapsulator Agents alter the chemical makeup of plain water droplets, leading to several advantages.

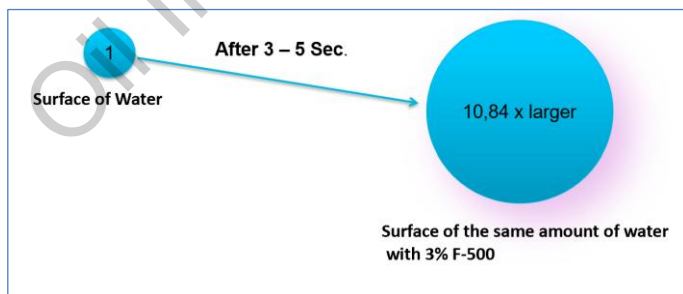
Encapsulator Agents start with a large amphipathic molecule, containing both a hydrophilic polar head and hydrophobic non-polar tail that act independently.



When mixed with water, these molecules will instantly form billions of micro spherical micelles as non-polar tails turn inward, avoiding water at all costs.



Encapsulator Agents immediately reduce the surface tension of water. This makes droplets smaller, creating more surface area to rapidly absorb heat and better penetrate the pores of solid material.



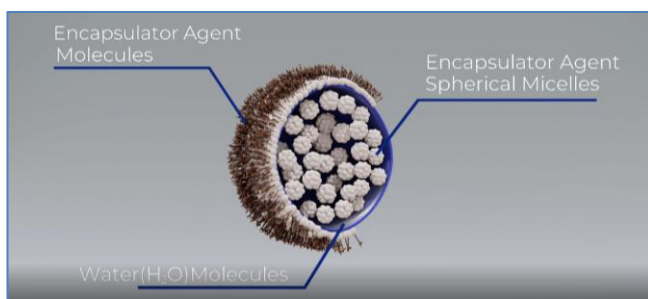
As Encapsulator Agent (EA) droplets leave the nozzle, non-polar tails near the surface turn outward, forming a protective “skin” around each droplet. This happens instantly, leaving a multitude of spherical micelles within each EA droplet, changing the EA droplet’s heat reduction mechanism.



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This enables an EA droplet to absorb significantly more heat than a plain water droplet. Thermal energy is driven inward towards the countless spherical micelles inside. A plain water droplet lacks the ability to absorb heat three -dimensionally and therefore generates scalding steam. This feature allows Encapsulator Agents to extinguish fires more efficiently than plain water and gives them the ability to safely mitigate fires that plain water cannot.

In addition to this, the non-polar tail of EA molecule completely wraps around / encapsulates any organic molecule present in the fuel and by-products of fire like smoke and off-gases. By encapsulating the fuel, it makes it chemically unavailable for further reduction and thus extinguishing fire.

**ANNEXURE-XI****MAJOR CHANGES IN THE REVISED EDITION****2.0 SCOPE**

Petrochemical complexes are added.

3.19 PETROCHEMICAL COMPLEX

Petrochemical complexes are normally petroleum or other hydrocarbon processing industries where chemical products or derivatives are derived from petroleum & focus to produce industrial & consumer products.

HDPE, LLDPE, DFCU, Butene, PET, PVC, etc are some process blocks in a petrochemical complex.

4.2.1 FIXED WATER SPRAY ON STORAGE TANKS

Class 'A' Petroleum in above ground tanks (floating or fixed roof) shall have fixed water spray system, having capacity larger than 190 m³ or diameter larger than 6 m.

4.2.2 SEMI-FIXED FOAM SYSTEM FOR STORAGE TANKS

- Fixed roof tanks storing Class 'A' and class 'B' petroleum products, of diameter larger than 9 m.
- Note: Care should be taken in applying portable foam streams to high-viscosity materials heated above 93.3°C (200°F). Good judgment should be used in applying foam to tanks containing hot oils, burning asphalts, or burning liquids that have a boiling point above the boiling point of water. Although the comparatively low water content of foams can beneficially cool such fuels at a slow rate, it can also cause violent frothing and “slop-over” of the tank’s contents.

4.2.4 AUTOMATIC WATER SPRAY FOR STORAGE AND HANDLING OF LPG /C4 AND LIGHTER ENDS/ HYDROGEN :

Automatic Water Spray shall be provided for the following areas:

- LPG/C4 and lighter ends and hydrogen, pressure storage vessels shall be provided with automatic water spray system.
- LPG bottling stations, C4 & Lighter ends - loading or unloading gantries, pumps, and compressors.
- Hydrogen compressors associated with storage facilities.

4.2.5 WATER SPRAY SYSTEM IN PROCESS UNIT

- Automatic water spray system for pumps handling LPG or hydrocarbons above auto-ignition temperature.
- Process compressors of C4 or lighter and Hydrogen.



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- Water spray rings for columns of height more than 45 M (above grade level) shall be provided. As an alternate, remote operated elevated monitors (at least two, positioned diagonally opposite sides) shall be installed at suitable elevation or high-capacity trolley mounted monitors shall be available, such that they should cover topmost potential source of leakage of column.

4.2.7 WATER SPRINKLER SYSTEM

A water sprinkler system should be provided in polymer extruder buildings, palletizing building, polymer bagging & its warehouse of a petrochemical complex excluding water reactive/sensitive materials storage area. Car parking in the basement of any refinery or petrochemical complex should also be protected with a sprinkler system. For application density & other design aspects of any fire sprinkler system, NFPA-13/IS 15105 may be referred.

4.2.8 WATER CURTAIN SYSTEM FOR NAPHTHA & GAS CRACKER FURNACES

A water curtain system shall be installed to prevent the ingress of hydrocarbon gases to cracker furnaces from within the Process Unit. The water curtain shall be provided in cases where hydrocarbon gases are being handled in the vicinity of the cracker furnace in the process unit. The water curtain shall be actuated using the Hydrocarbon (HC) detectors placed in the vicinity of the cracker furnace. The water curtain shall be designed as per NFPA 13 at a design flow rate of 37 lpm/meter. The water curtain coverage should extend a minimum of 1 m above the flame box of a furnace.

4.2.9 CLEAN AGENT FIRE SUPPRESSION SYSTEM FOR CONTROL ROOM, INSTRUMENT PANEL ROOM & SATELLITE RACK ROOM (SRR)

Clean Agent System along with all equipment, software for hydraulic calculations and corresponding design concentration of clean agent shall be listed or approved by national or international standards such as BIS/UL/FM/VdS/LPCB etc. Cylinder along with cylinder valve assembly shall also be PESO approved."

4.2.12 FIRE PROTECTION FOR COAL HANDLING AND CONVEYOR SYSTEM

Refinery/Petchem facilities where Coal based Captive Power plant available for power and steam generation.

Following Fire protection system shall be installed for the protection of facilities:

- Fire Hydrants, wet risers, and Hose reels
- Fire water Monitors – For structures having height more than 15 m.
- Automatic High Velocity Water spray system for Boiler burner fronts
- Manual medium velocity water spray system with proportioner to mix water with suitable UL approved wetting agent for Coal silos.
- Water sprinkler system for open type coke/coal conveyor.
- Suitable fire detection system.



4.2.13 FIRE PROTECTION FOR STORAGE / HANDLING OF FLAMMABLE CRYOGENIC LIQUIDS

Cryogenic liquids are liquefied gases that are maintained at very low temperatures ($< -150^{\circ}\text{C}$). These liquids are gases at ambient temperatures and pressures. Since these liquids are extremely cold and can expand to very large volumes of gases, they are a safety hazard.

When these liquids leak or catch fire, we need to be ready to handle an emergency. Fire protection measures required for the same are as below.

- Portable Fire Extinguisher Install as per Section 11.1
- Fire water network / Hydrants & Monitors: Install as per section 5.6.6 & 5.6.7
- Foam system: Portable / fixed High expansion foam generator shall be provided for protection of :
 - DWST (double walled storage tank) dykes for handling spills.
 - Underground (Pit) liquid transfer pumps.

5.0 FIRE WATER SYSTEM

Water is a universal fire extinguishing agent and meets the majority of ideal properties required for fire extinguishment.

The ideal fire extinguishing agent should have the following characteristics: -

- a. It must be non-flammable
- b. It should have a high heat of vaporization
- d. It should be mobile liquid at normal ambient temperature.
- e. It should be non-toxic and should not decompose to form toxic products.

Thus, water is the best available option and can be used for fire extinguishment, fire control, cooling of equipment and protection of equipment as well as protection of personnel from heat radiation. Water should be used such as straight jet, water fog, water mist, water curtain, water spray, deluge/sprinkler, for foam making etc.

The firewater system shall comprise of fire water storage, fire water pumps and distribution piping network along with hydrants and monitors, as the main components.

5.2 HEADER PRESSURE

The fire water system shall be designed for a minimum residual pressure of 7.0 kg/cm^2 at the hydraulically remotest point of application at grade level and minimum residual pressure of 5.25 kg/cm^2 at landing valve at elevated structure on designed flow rate at that point.

The fire water network shall be kept pressurized at minimum 7.0 kg/cm^2 at all the time.

5.4.3 STANDBY PUMPS

In case total number working pumps are up to 2, standby pumps shall be at least one. In case number of working pumps are 3 to 4, the number of standby pumps shall be at least 2. For more than 4 working pumps, the number of standby pumps shall be at least 3.

5.4.4 JOCKEY PUMPS

The capacity of jockey pumps shall be 1% minimum of the design fire water rate. Its head shall be higher than the main fire water pumps. Auto cut-in/cut-off facility should be provided for jockey pumps.



5.5.3 PROTECTION FOR UNDERGROUND PIPELINES

Where the pipes are laid underground the following protections shall be provided:

The mains shall be provided with protection against soil corrosion by suitable coating/wrapping. Where pipes are coming out/going in from/into the ground, the underground anticorrosion coating on the pipe will continue for a length of at least 300 mm above ground. In concrete paved areas, removable paver block shall be provided over the buried fire water lines for ease of maintenance without cutting of pavement.

5.6.4 FIXED REMOTE / MANUAL OPERATED HIGH VOLUME LONG RANGE WATER CUM FOAM MONITORS

Below criteria should be followed for provision of water cum foam monitors in tank farm area.

a) The number and capacity of monitors shall be provided in such a way that cumulative foam application rate from the designated monitors meet the application rate of 9.75 lpm/m² for total burning surface of the affected tank to extinguish the full surface tank fire. Additional monitors shall be provided in such a way that each tank is in the coverage area of at least two monitors. Due consideration has been given to compensate for the losses (50%) due to wind and other factors.

b) Monitors should be located to achieve the foam discharge preferably close to tank center from upwind direction of tank on fire.

d) A suitable connections for water supply to portable trailer mounted HVLR should be made available.

The basic water cum foam monitors shall be complying to the National or international standards such as BIS, UL and FM etc or certified by the approved Third-Party Inspection (TPI) agency of the entity. The electrical or hydraulic remote-control mechanism shall be conforming to Hazardous Area Classification.

5.8.2 WATER SPRAY APPLICATION RATES

The following water spray application rates are recommended for general guidance. These rates should be reviewed on a case-to-case basis and increased, if required. While calculating the water rates for spray application for cases other than tanks/vessels, the area should be divided into suitable segments so that maximum water requirement for spray application should generally not exceed 1200 m³/hr.

Application Area

Atmospheric storage

Tanks

Water application rate

3 lpm/m² of tank shell area for tank on fire

3 lpm/m² of tank shell area for exposure protection for tanks located within in the same dyke area.

6.2 FOAM CONVEYING SYSTEMS

The system consists of an adequate water supply, supply of foam concentrate, suitable proportioning equipment, a proper piping system, foam makers and discharge devices designed to adequately distribute the foam over the hazard.

Conventional systems are of the open outlet type in which foam discharges from all outlets at the same time, covering the entire hazard within the confines of the system. It shall be ensured that continuous foam concentrate supply from source to foam equipment is available.



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6.4 FIXED ROOF TANK PROTECTION USING FOAM

- i) Tanks should be provided with foam discharge outlets/ vapour seal chambers as indicated below:

Tank Diameter in M.	Minimum number of foam discharge outlet
≥9 upto 20	2
>20 upto 25	3
>25 upto 30	4
>30 upto 35	5
>35 upto 40	6
>40 upto 45	8
>45 upto 50	10
Plus 1 outlet for each additional 465 m ² of area	

6.6 FULL SURFACE STORAGE TANK FIREFIGHTING & LOGISTICS

Large Diameter Full Surface Tank Fire Fighting:

Full surface Tank Fire Scenario:

Part A: Tanks diameter up to 45 meter

It is generally experienced that full surface fire in smaller diameter petroleum storage tank up to 40-meter diameter are relatively easy to extinguish with reliable resources and competent manpower.

This shall be achieved by designing dyke foam monitors of adequate capacity / foam tender roof monitors. Un-interrupted foam supply shall be ensured by any reliable means for a continuous period of 65 minutes with extended coverage for an additional 30 minutes.

Part B: Tanks diameter larger than 45 meters

Large diameter full surface tank fire is a challenge and requires significant preparation in terms of resources and well proven strategy.

Successful extinguishment of a large diameter petroleum storage tank full surface fire begins with pre-fire planning and therefore most of the following information shall be considered. Entity to prepare the pre-fire plan for largest diameter tank at site and list out the resources required. The priority shall be for rescue and evacuation of personnel before attempting extinguishment.

The manpower requirement to tackle a major tank fire may vary depending upon the type, location and nature of the fire, method of extinguishment and the availability of trained personnel. This shall be determined while developing the pre-fire plans.

Resources Requirement:

Below listed major resources shall be made available at site as a design requirement and periodically these resources should be tested and maintained in ready to use condition. The critical aspect of successful tank fire extinguishment is a stable uninterrupted foam delivery in required quantity through aerial (over the top) application. Foam discharge thru large volume monitors



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should reach close to tank center for effective foam blanketing over burning surface. Many successful full-surface tank fire fighting are achieved by way of throwing foam through 2-3 large volume foam monitors from upwind side of the burning tank. Multiple direction foam discharge over a burning tank may not give a satisfactory result.

A typical list of major Trolley /portable / mobile equipment for full surface tank fire is given below.

- a. High Volume Long Range (HVLR) foam monitors
- b. Water monitors
- c. Foam pumping/ proportioning appliances
- d. Foam concentrate tankers or containers (Tote containers)
- e. Fire hose including large diameter/ capacity hose and Water supplies

a. High Volume Long Range (HVLR) Foam Monitors

The foam monitor capacity and range are important factors for fire extinguishment. Foam streams have to be such that the bulk of the discharge reaches the tank liquid surface. It is recognized that the best method of application is to project foam with the wind behind the stream and not against. However, there have been, and will be situations where cross winds or a variable breeze causes destructions in stream range and these needs to be considered. Therefore, it is important to use large capacity trailer mobile foam monitors for tank fire extinguishment, which give the versatility for placement considering the prevailing wind direction so that the bulk of the output reaches the tank liquid surface without affecting the stream range. Number and capacity of large volume foam monitors should be decided based on Pre-fire plan for largest diameter tank fire. Foam monitors are available ranging from 1000 to 12000 GPM.

b. Water Monitors

Water monitors are necessary for cooling adjacent tanks affected by radiant heat or direct flame impingement. Water cooling of the tank on fire is not normally necessary except to assist foam blanket sealing against a hot tank wall. Cooling may be attempted if the water spray system is intact and operable for uniform cooling.

c. Foam proportioning appliances

An uninterrupted foam supply with required induction rate matching with foam monitors discharge for the total duration of foam application is necessary for extinguishment. It is very essential that the selected foam proportioning system is suitable for the logistical challenges involved in bulk foam concentrate handling. It is possible to supply foam to the foam monitors either using foam pumpers, or foam fire tenders or through foam pick up of a monitor.

d. Foam concentrate and supply considerations

Availability of required quantity of foam concentrate at emergency site shall be ensured before commencing foam application. Once foam application commences, it shall be sustained for a minimum period of 65 minutes @ 9.75 lpm/m². It is recommended that bulk storage is provided in foam nuser trucks, elevated bulk storage tanks or 1000-liter Tote containers on mobile trailers / trucks. Conventionally 3% foam concentrates are used for tank fire fighting, however 1% foam



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concentrate can be considered for better logistics management. It is highly recommended to keep the total quantity of foam as a separate stock to be mobilized only during tank fire scenario.

It is experienced that foam losses during actual firefighting is high around 50-60 % during heavy wind or adverse weather condition or flame draft from the burning tank. Safety allowance of additional 50% has been included in the above application rate.

Foam compatibility is an important factor. Mixing foam concentrates of different types is not recommended and can destroy foam blanket over the burning surface. Moreover, foam concentrate should be compatible with associated resources being used for firefighting.

Suggested options for Foam concentrate supply-

- Foam Nurser / Foam Transport Vehicle / Foam Tender
- Foam Tote on Trailer with common discharge manifold

e. Fire Hose and Water supply

Wherever the full surface firefighting strategy is adopted with mobile trailer mounted high-capacity water cum foam monitors, large capacity hoses for feeding water to the monitor should be selected to avoid losses / pressure drop during actual operation. The sizes of large capacity hoses vary from 4" to 12". Conventional 63mm fire hoses may be used for portable water monitors for cooling and foam proportioning system. Power driven device may be used for quick laying of hoses.

Fire water from available fire water network / large capacity pumps should be used to feed large diameter hoses for monitors. Suitable feeding arrangement should be made available for this water supplies.

Note: For foam calculation refer Annexure-VI.

6.8 DURATION OF FOAM DISCHARGE

For floating roof tanks containing liquid hydrocarbons, foam solution delivery rate shall be at least 12 lpm/m² of seal area with foam dam height of 600 mm of the tank to be protected. In case of floating roof sinking, the rate considered should be 9.75 lpm/m² of liquid surface areas.

The equipment shall be capable of providing primary protection at the specified delivery rates for the following minimum period of time.

- i) Tanks containing liquid hydrocarbons - Class 'C' Petroleum - 30 minutes.
- ii) Tanks containing Class 'A' & 'B' Petroleum or liquids heated above their flash points - 65 minutes.
- iii) Where the system's primary purpose is for spill fire protection - 30 minutes.

Note:- It is advised to keep foam compound stock for extended 30 minutes for full surface tank fire scenario.

6.10 FOAM COMPOUND STORAGE

- a. Foam compound should be stored in containers of suitable capacity, in case of protein, fluoro protein or AFFF or AR-AFFF or FFFP. Alcohol Resistant Foam(AR-AFFF) shall be used for handling methanol/ ethanol fires. Alcohol Resistant Foam (AR-AFFF) can be used for all type of



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hydrocarbon fire and polar solvent fire effectively. Foam compound can also be stored in overhead storage tank of suitable capacity for quick filling of foam tender / nurser during emergency.

Cautionary: Alcohol Resistant Foam (AR-AFFF) is preferred for Ethanol Blended Petrol (EBP) fire.

8.0 BATTERY FIRE

From consumer electronics and electric vehicles to industrial equipment and energy storage systems, lithium-ion batteries can be found everywhere. Due to the materials utilized to manufacture these batteries, the scene of a lithium-ion battery fire can best be compared to a hazmat (hazardous material) incident. Although extinguishing fire is a primary concern, flammability, explosivity and toxicity must be addressed to ensure proper protection and prevent re-ignition.

In a cell or battery, ignition occurs when lithium-ion batteries enter a chemical process called thermal runaway. Cells produce heat and emit explosive and toxic gases, developing a self-propelling loop that leads to thermal runaway propagation. This chain reaction will continue to impact adjacent cells until thermal runaway propagation is halted. Confined spaces such as server rooms, automobiles, aircraft, spacecrafts, shipping containers, energy storage systems and more can pose an increased risk.

During the battery fire, a flammable electrolyte used within cells catches fire. Thus, fire starts inside a cell which makes it difficult to reach. All the conventional firefighting media, including plain water, works at the macro / surface level to remove heat. In the case of battery fire, due to deep seated nature of fire, the effectiveness of conventional firefighting media is very less and thus they are required in humongous quantity to achieve extinguishment.

In many battery fires, one of the byproducts of the oxidation (burning) chemical process is oxygen, thus the fire is fed by creating its own oxygen. Hence, any firefighting media / techniques that works on a principle of oxygen depletion find limited success.

8.1 FIRE PROTECTION

For fire protection of Li ion battery, Encapsulator Agent (EA) technology should be used in line with NFPA 18A. (For details refer Annexure-X)

10.1 RECOMMENDED USE:

A dry chemical powder extinguishing system should effectively be used on following hazards.

- Pyrophoric materials and alkyl metal compounds(Such as TEAL,TEB).

11.0 MODERNIZATION PLAN FOR FIREFIGHTING EQUIPMENT:

As a part of modernization of firefighting equipment, following may be explored based on its suitability As a part of modernization of firefighting equipment, following may be explored in addition to the OISD-GDN-115 based on its suitability to hazards, long term maintainability, reliability and certification.



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a) Dual Agent Nozzle

The nozzle is designed to throw liquid (foam solution / water) and DCP simultaneously. The liquid forms a hollow jet and the DCP is inside the hollow space. Application wise DCP provides quick knock down of flames and the liquid provides cooling. Also, DCP can be thrown at a longer distance than conventional DCP hand nozzle or monitor.

b) Fire extinguisher of HDPE / Composite material body:

This will ensure less weight and claimed to have longer shelf life. This will be more useful in corrosion prone area like offshore.

13.1 FIRE TENDERS:

The exact number of *fire tenders* shall be higher of the items (a) or (b):

(a) The quantities fixed up in each case based on two simultaneous major fire taking into consideration the size, location of the plant and statutory requirements.

(b) The quantities indicated below.

i) 3nos. of foam tenders out of which two are for firefighting, one for spill /standby. The foam tender should have a foam tank capacity of at least 3000 Liter and the pump capacity of minimum 4000 lpm at 8.5 kg/cm².

ii) DCP tender(s) having aggregate 4000 kg capacity, with Nitrogen as expellant gas. These are required for fighting LPG/Gas fire.

A monitor should have a variable throw of 15/25/40 kg/sec. The throw of the monitor shall be 40 to 50 M for the DCP charge.

Minimum 1 numbers of Foam Nurser with foam compound tank of minimum 8000 litre capacity, with suitable pump for foam transfer.

For Gas Processing Units a minimum of 2 nos. of multipurpose fire tender shall be provided.

Note:- For small petrochemical follow (a) clause of this section.

13.2 OTHER MOBILE EQUIPMENT

13.2.1 In addition to fixed monitors provided in the tank farm, following additional mobile equipment's should be provided:

- i) Minimum 2 nos. of foam tank trailers preferably with field adjustable variable flow water cum foam monitors having foam tank capacity of 500-1000 liters and monitors capacity of minimum 1000 GPM (228 m³/hr) and which are listed or approved by standards such as BIS, UL, and FM etc.
- ii) Minimum 2 nos. of Trolley mounted water cum foam monitors of capacity of minimum 2000 GPM (456 m³/hr) with field adjustable variable flow and which are listed or approved by national or international standards such as BIS, UL and FM etc. Foam induction to the monitor shall be possible from minimum 60M distance from the monitor.



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- iii) 1 to 2 nos. of *portable*/ trailer fire pumps of capacity ranging from 1800 to 2250 lpm at discharge pressure of 7 kg/cm².

13.2.2 For Gas processing plants, in addition to fixed monitors provided in the tank farm, following additional mobile equipment should be provided; namely:-

(i) Minimum 2 nos. of Trolley mounted water cum foam monitors of capacity of minimum 2000 GPM (456 m³/hr) with field adjustable variable flow which are listed or approved by national or international standards such as BIS, UL and FM etc. Foam induction to the monitor shall be possible from a minimum 60 m distance from the monitor.

(ii) 1 to 2 numbers. of portable monitor of capacity ranging from 1800 to 2250 lpm at discharge pressure of 7 kg/cm².

15.1.4 TOXIC GAS DETECTION:

- Toxic gas detectors shall be installed if the risk is applicable. Toxic gas detectors shall be mounted considering the prevailing wind direction and wind speed.
- Additionally, Toxic gas sensors shall be installed to HVAC air intakes to buildings that could be affected in the event of flammable or toxic release in the surrounding area.



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**1ST FUNCTIONAL COMMITTEE
(March 1989)**

NAME	ORGANIZATION
LEADER	
M.M. Kapoor (w.e.f. 1.1.95)	Engineers India Limited
R.P. Bhatla (upto 31.12.94)	Engineers India Limited
MEMBERS	
A. A. Raichur	Hindustan Petroleum Corporation Limited
S. Neelkanthan	Madras Refineries Limited
P.J. Joshuva	Cochin Refineries Limited
Sunil Kumar	Indian Oil Corporation Limited
V.P. Vaidhya	Hindustan Petroleum Corporation Limited
S.N. Mukherjee	Bharat Petroleum Corporation Limited
A.K. Das	Bongaigaon Refineries & Petrochemicals Ltd.
R.P. Saxena	Oil & Natural Gas Corporation Ltd.
H.K. B. Singh	IBP Co. Ltd.
MEMBER CO-ORDINATOR	
Vijay M. Ranalkar	OISD



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**2nd FUNCTIONAL COMMITTEE
(March 2007)**

NAME	ORGANIZATION
LEADER	
Arvind Kumar	Engineers India Limited
MEMBERS	
A. A. Raichur	Hindustan Petroleum Corporation Limited, Mumbai.
S. P. Garg	GAIL (India) Limited, Pata (UP).
G. B. Tolmore	Indian Oil Corporation Limited, New Delhi.
G. C. Kundu	Indian Oil Corporation Limited, Panipat.
S.N. Mukherjee	Bharat Petroleum Corporation Limited. Mumbai.
P. K. Bora	Bongaigaon Refineries & Petrochemicals Ltd, Bongaingaon.
A.K. Das	Kochi Refineries Limited, Kochi.
K. V. Singh	Hindustan Petroleum Corporation Limited, Vizag.
J.P.K. Hapat	Chennai Petroleum Corporation Limited, Chennai.
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MEMBER CO-ORDINATOR	
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